

ZS-Q16

Single-Outcome Selection as a Z-Spin-Mediated Self-Referential Measurement Closure: Weak Sufficiency from the $\dim(Z)=2$ Holographic Bottleneck and the Non-Epistemic Localization of the Residual

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Locked inputs: (A, Q, $\dim(Z)$) = (35/437, 11, 2) **LOCKED**

v2.5 Release Banner. v2.5 is strictly additive over v1.0–v2.4 (no prior content deleted). It adds §22, which examines a proposed Spin-Geometric Realization route to close Theorem Q16.C2’s geometric residual (the embedding of $\partial_{\text{op}} X$ as a codim-1 spin boundary of the 3D X-bulk, the $SO(3)$ -frame source of the 4π closure). Honest outcome: this route DECOMPOSES the residual but does NOT close it. Theorem Q16.S0 (Operational-to-Regge, DERIVED-under-Regge) inherits — does not derive — the Regge boundary structure of ZS-Q6. Theorem Q16.S1 (Regge Boundary Spin-Lift, IMPORTED) supplies $\text{Spin}(3) = \text{SU}(2) \Rightarrow 4\pi$, but only after assuming the codim-1 embedding (a 2D Regge boundary OF a 3D bulk), and on a bare surface the spin structure is a CHOICE forced only by that embedding (ZS-Q12V Thm q12.dim2). Hence “ReggeSpinBoundary” is logically equivalent to the geometric embedding $Z=\partial X$ — a clarifying decomposition into three named sub-conditions (Regge realization, normal-bundle reconstruction, oriented-frame closure), NOT a reduction to anything more primitive. Theorem Q16.C2 is therefore NOT promoted; it remains DERIVED-CONDITIONAL, its condition now explicitly decomposed rather than closed. The one genuine new lead — reconstructing the normal bundle from the X-side record flow (an information→geometry bridge) — is registered as O-Q16.12, which connects to the external emergent-spacetime program. New: Theorems Q16.S0–S1, Corollary Q16.S2, NC-Q16.14, O-Q16.12, and two references (Regge 1961; Milnor–Stasheff 1974). The §22 spin-lift facts ($\text{Spin}(3) = \text{SU}(2)$) are already verified in zs_q16_verify Cat H. Zero new free parameters; (A, Q, $\dim Z$) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

Verification Summary

Verification: 27/27 PASS across Categories A–E (zs_q16_outcome_mc_v1_0.py, canonical seed 20260604) | Cross-paper consistency 9/9 PASS | **Zero Free Parameters** | A = 35/437, Q = 11, (Z, X, Y) = (2, 3, 6) sole geometric inputs | Anti-numerology: the per-transit information ceiling is shown to equal $\ln(\dim Z)$ to machine precision and to be dimensionally locked across $d = 2 \dots 6$ (ratio = 1.000000), i.e. theorem-fixed (Holevo $\times$ ZS-Q7), not a fitted coincidence | Falsification gates: 6 pre-registered (F-Q16.1–6), multi-layered (mathematical / simulation / observational).

§0. Abstract

The Tomaz–Mattos–Barbatti outcome problem — why a single definite result is obtained on a given measurement run after decoherence has fixed the pointer basis — was registered in ZS-F19 v2.0 as **NON-CLAIM** NC-F19.7: the Z-Spin rank-2 channel supplies a *kinematic* binary-outcome structure but

not a *dynamical* single-outcome selection. This paper resolves the status of NC-F19.7 by decomposing the outcome problem into three nested levels and locating exactly where the Z-Spin geometry closes it and where it cannot. We first separate the ZS-Q1 §3.4 dephasing unraveling (a no-record process that produces ensemble decoherence only) from the ZS-Q12 Q12.A self-referential measurement closure (a record-keeping process that localizes each trajectory). The ZS-Q14 Boundary Spin-Lift ($\chi_Z = -1$, process-level witness $\tilde{u}_Z \neq f(\Lambda)$) selects the latter; the ZS-F1 vanishing block $L_{XY} \equiv 0$ with the rank-2 ZS-Q7 bottleneck guarantees that the single mediating channel observes all cross-sector information, so the Maassen–Kümmerer purification condition (absence of dark subspaces) is met geometrically. Importing the rigorous convergence theorems for quantum non-demolition trajectories (Bauer–Bernard; Bauer–Benoist–Bernard; Benoist–Pellegrini), reinterpreted through the Z-Spin dictionary with the *i*-tetration fixed point z^* as the attractor, we promote **weak sufficiency** — almost-sure collapse to a single eigenstate with Born statistics — from NC/OPEN to **DERIVED-CONDITIONAL**. The collapse proceeds at a per-Z-transit information rate bounded by $\ln 2 = \ln(\dim Z)$, the Holevo ceiling identified with the ZS-Q7 channel capacity. For **strong sufficiency** — the elimination of the per-run randomness — we report a sharp negative structural result: the estimation-stability theorem (Benoist–Pellegrini), verified here at 2000/2000 prior-agreement, shows the per-run outcome is a function of the measurement record and is **non-epistemic** (no refinement of state knowledge alters it). The residual randomness is therefore not a knowledge gap but the irreducible two-dimensional Z-innovation; strong sufficiency remains **OPEN**, now structurally localized. The absolute-strongest form (the ontology of the definite record) is interpretation-dependent and retained as **NON-CLAIM**; the corpus consciousness firewall (NC-Q7.4 / NC-A7.6 / NC-F10.3 / NC-F11.1 / NC-F19.1; gates F-F11.7' / F-F19.6) is untouched. Zero new free parameters.

Keywords: outcome problem; single-outcome selection; quantum non-demolition; Belavkin filter; quantum trajectory; purification; estimation stability; non-epistemic randomness; Z-bottleneck; i-tetration fixed point; $\dim(Z) = 2$; zero free parameters.

Epistemic Status Legend

All claims in this paper carry exactly one of the following tags, reproduced from the Z-Spin corpus convention (ZS-F0 v1.0(R); ZS-F19 v2.0) for cross-paper compatibility. Ad hoc qualifiers outside this set are not permitted.

Status	Definition
PROVEN	Mathematical theorem with complete proof under stated definitions, or numerical verification at machine precision ($\leq 10^{-10}$ residual).
DERIVED	Quantitative consequence of PROVEN items combined with the Z-Spin action, with zero free parameters beyond $A = 35/437$.
DERIVED-CONDITIONAL	Derived from Z-Spin axioms conditional on a stated, explicitly tracked assumption or upstream theorem.
IMPORTED	Result proved externally and used here without re-proof; full citation given.
VERIFIED	Numerical / computational confirmation at stated precision strengthening a DERIVED claim.
TESTABLE	Quantitative prediction with a pre-registered falsification protocol.
HYPOTHESIS-strong	Motivated conjecture with multiple independent structural anchors; promotion path documented.
OBSERVATION	Empirical / structural regularity, anti-numerology controlled; structural origin pending.

Status	Definition
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework's scope.
OPEN	Recognized gap honestly registered for future work; promotion path may exist.
LOCKED	Core constant fixed upstream; no downstream paper may modify.

§1. Introduction

§1.1 The residual outcome problem

Environmental decoherence explains the suppression of interference and the selection of a preferred (pointer) basis, but it is widely recognized that it does not, by itself, explain why a *single* definite outcome is realized on an individual run [11, 14]. Tomaz, Mattos and Barbatti [15] sharpen this as the *outcome problem*: the residual question that survives decoherence. In the Z-Spin corpus this was registered with full honesty as **NON-CLAIM** NC-F19.7 (ZS-F19 v2.0 §6.4): the vanishing X–Y block $L_{XY} \equiv 0$ (ZS-F1 v1.0, **PROVEN**) and the rank-2 Kraus dilation (ZS-Q1 v1.0 §3.3; ZS-Q7 v1.0 Theorem 2, **DERIVED**) force every cross-sector measurement to factor through a two-dimensional channel, giving a binary-outcome space with Born weights — a *kinematic* production mechanism — but they do not address the *dynamical* selection of one outcome on one run.

This paper does not overturn NC-F19.7; it **resolves its status** by separating three questions that the single phrase “the outcome problem” conflates, and determining, for each, whether the Z-Spin geometry plus rigorous external mathematics closes it.

§1.2 Two unravelings of the same master equation

The pivot is a distinction internal to the corpus. ZS-Q1 §3.4 establishes an emergent Lindblad equation through a stochastic Schrödinger equation with seam-parity noise,

$$d|\Psi\rangle = [-(i/\hbar)H - \gamma/2] |\Psi\rangle dt + \sqrt{\gamma} \sigma_z |\Psi\rangle dW(t), \quad \Gamma = 2\gamma = 2A(\Delta E/\hbar)^2, \quad (1)$$

whose ensemble average obeys $dp/dt = -(i/\hbar)[H, \rho] + \Gamma(\sigma_z \rho \sigma_z - \rho)$. Because the Lindblad operator σ_z is Hermitian, the **populations are conserved along each trajectory**: this unraveling randomizes only the relative phase and produces ensemble decoherence with no per-trajectory localization. It is exactly the “necessary but not sufficient” content the outcome problem flags. ZS-Q12 v4.0 §4.3 (Q12.A), by contrast, invokes the Belavkin / stochastic-master-equation dynamics, which is *nonlinear* because Born-rule normalization feeds the conditional state into its own generator; wave-function collapse there is convergence to a fixed point of this self-referential map, realized at unit self-reference gain. The same generator, two unravelings: one keeps no record and does not select; one keeps a record and does.

§1.3 What this paper establishes

(i) Which unraveling is physical is selected by the ZS-Q14 Boundary Spin-Lift (§4). (ii) The purification condition that makes collapse generic is met geometrically by $L_{XY} \equiv 0$ plus rank-2 (§4–§5). (iii) Weak sufficiency is **DERIVED-CONDITIONAL** via imported non-demolition convergence theorems reinterpreted with z^* (§5). (iv) The collapse rate is ceilinged by $\ln(\dim Z) = \ln 2$, a theorem-fixed value (§6). (v) The strong-sufficiency residual is **OPEN** but **non-epistemic and structurally localized** (§7). (vi) The absolute-strongest form is **NON-CLAIM** (§7.7). This paper is *not* a theory of consciousness, a

new collapse model with stochastic parameters (cf. GRW/CSL [12]), or a re-derivation of the Born rule (imported unchanged from ZS-Q1).

§2. Locked Inputs

All quantities are inherited from prior corpus papers; no new constant is introduced.

Table 2.1. Locked and imported inputs to ZS-Q16. Bold symbols are core constants.

Symbol	Value / meaning	Status	Source
A	35/437 (geometric impedance)	LOCKED	ZS-F2 v1.0
Q	11 (register dimension)	LOCKED	ZS-F5 v1.0
(Z, X, Y)	(2, 3, 6) sector dimensions	PROVEN	ZS-F5 v1.0
dim(Z)	$2 = j = 1/2$ spinor boundary	PROVEN	ZS-M3 v1.0
$L_{XY} \equiv 0$	vanishing X–Y block (Z-mediation)	PROVEN	ZS-F1 v1.0
$\text{rank}(T_{XY}) \leq 2$	Z-bottleneck; $\text{cap} \leq \ln 2$	DERIVED	ZS-Q7 v1.0 Thm 2
w_Y	$\text{dim}(Y)/Q = 6/11$ (Born weight)	PROVEN	ZS-Q1 v1.0 §4.2
z^*	$0.43828 + 0.36059 i$; $ f'(z^*) = 0.892$	PROVEN	ZS-M1 v1.0
$\lambda = (i\pi/2)z^*$	Wilson eigenvalue; $ \lambda = 0.8915$	DERIVED-COND.	ZS-Q12 v4.0
$\lambda_2 = 2A/Q$	0.014562 (Lindblad coefficient)	DERIVED	ZS-Q15 v1.2
$\chi_Z = -1$	spinor lift class ($2\pi \mapsto -1$)	DERIVED-COND.	ZS-Q14 v1.4 Lem 5.1

§3. The Three-Level Decomposition of the Outcome Problem

We separate three questions that NC-F19.7 had bundled. Distinguishing them is the central methodological move of the paper: the Z-Spin geometry closes the first, structurally localizes the second, and bounds out the third.

Table 3.1. The three nested levels of the single-outcome question and their Z-Spin status.

Level	Precise question	This paper's verdict
Weak	On a given run, is a single definite eigenstate realized, with frequencies equal to the Born weights?	DERIVED-CONDITIONAL (§5): yes, via imported QND convergence theorems + z^* attractor + purification from $L_{XY} \equiv 0$.
Strong	Is the per-run randomness (which outcome) eliminable — reducible to a deterministic or fuller description?	OPEN, non-epistemic (§7): the outcome is a function of the record; estimation stability rules out reduction via state knowledge. Residual = $\text{dim}(Z)=2$ innovation.
Absolute	Why does a definite record exist at all (single-world fact vs. branching)?	NON-CLAIM (§7.7): interpretation-dependent; not closable by any measurement dynamics, Z-Spin included.

§4. The Z-Channel as a Quantum Non-Demolition Binary Measurement

§4.1 Pointer, channel, and the no-dark-subspace property

The measured observable is the Z_2 seam parity σ_z (ZS-Q1 §2); its eigenbasis is the pointer basis, and the binary outcome space $\{|0\rangle, |1\rangle\}$ is the rank-2 = $\dim(Z) = 2$ outcome space of NC-F19.7. By $L_{XY} \equiv 0$ the X-sector communicates with the Y-environment *only* through this single two-dimensional Z-channel (ZS-Q1 §3; ZS-Q7 §3). The Maassen–Kümmerer purification theorem [6, 7] states that a continuously monitored trajectory almost-surely purifies *unless* the dynamics hits a *dark subspace* — a subspace from which no information leaks to the observed environment. In Z-Spin there is no unobserved leak channel: the rank-2 Z-bottleneck is the unique conduit, so for the cross-sector pointer there is no dark subspace and the purification condition is met **geometrically**. This is the structural reason the collapse is generic rather than fine-tuned.

§4.2 Boundary Spin-Lift selects the record-keeping unraveling

A channel alone does not select an unraveling, just as a CPTP map does not select a Kraus lift (ZS-Q14 v1.4, Choi-Invisibility No-Go, **PROVEN**; non-identifiability, **PROVEN**). The Boundary Spin-Lift Lemma (ZS-Q14 Lemma 5.1) shows that a physical Z-boundary carrying the nontrivial spinor class of $SO(3)$ ($\dim Z = 2$, $2\pi \mapsto -I$) forces the half-angle connection and $\chi_Z = -1$, with a *process-level witness* $\tilde{u}_Z = F(V, G_Z, \varphi_Z) \neq f(\Lambda)$ — an observable carried by the dilation/comb beyond the channel. We read this as the structural statement that the Z-boundary **keeps a record**: the physical unraveling is the measurement (record-keeping) one of Q12.A, not the no-record dephasing of (1). The identification “process-level lift witness $\Rightarrow$ measurement unraveling” is a structural correspondence between two distinct non-uniqueness results (Kraus-lift vs. unraveling) and is carried at **HYPOTHESIS-strong** (gate F-Q16.3, promotion path O-Q16.3).

§5. Weak Sufficiency: Imported Convergence Theorems Reinterpreted

Given the measurement unraveling and the no-dark-subspace property, the following rigorous external theorems apply directly. They are **IMPORTED**; the Z-Spin contribution is the dictionary that places them on the geometric channel.

(T1) Bauer–Bernard [2] prove that repeated indirect quantum non-demolition measurements converge to wave-function collapse as predicted by the measurement postulate, with the convergence rate equal to the relative entropy of each indirect measurement. (T2) Bauer–Benoist–Bernard [3] take the continuous-time limit and show the pointer distribution satisfies the Born rule with exponential convergence. (T3) Benoist–Pellegrini [4] prove that the QND stochastic-master-equation solution converges, for large time, to a *random pure state* — the wave-packet reduction — “as if a direct von Neumann measurement had been performed at time 0,” with an exponential rate fixed via Girsanov. The necessary-and-sufficient condition for the SME to be a non-demolition measurement is that the pointer probabilities form a martingale. (T4) Maassen–Kümmerer [5, 6] supply the purification theorem invoked in §4.1.

The Z-Spin dictionary:

Table 5.1. Dictionary placing the imported non-demolition convergence theorems on the Z-channel.

Imported object (T1–T4)	Z-Spin reinterpretation
Probe on which a von Neumann measurement is made	Z-seam parity ($\dim Z = 2$ binary probe)
One indirect measurement	One Z-transit; information $\leq \ln 2 = \ln(\dim Z)$ (ZS-Q7)
Convergence to a random pure state	Attraction to the i-tetration fixed point z^* ($ f(z^*) = 0.892 < 1$; ZS-M1, ZS-Q12 Q12.A)

Imported object (T1–T4)	Z-Spin reinterpretation
Pointer probabilities are martingales (N&S)	Trajectory counterpart of the ZS-Q7 D_KL Lyapunov monotone decrease
No dark subspace (purification holds)	$L_{XY} \equiv 0$ + rank-2 single observed channel (geometric)
Born statistics of the limit	Projection weight $w_Y = \dim(Y)/Q = 6/11$ (ZS-Q1 §4, PROVEN)

Conclusion (weak sufficiency, **DERIVED-CONDITIONAL**). Conditional on (a) the Z-boundary being a physical spinor measurement interface ($\chi_Z = -1$; §4.2) and (b) the ZS-Q12 4π spin-closure axiom that fixes the z^* generator, the conditional state collapses almost surely to a single σ_z eigenstate with Born frequencies. This promotes the weak level of NC-F19.7 from NON-CLAIM/OPEN to **DERIVED-CONDITIONAL**. The promotion route is a theorem (T1–T4), not a numerical coincidence; it therefore does not require — and is not weakened by the absence of — an anti-numerology Monte Carlo (see §9).

§6. The Rate Ceiling: $\ln(\dim Z)$ from Holevo $\times$ ZS-Q7

Let the pointer-basis Shannon entropy be $H(q) = -\sum q_n \ln q_n$. The expected one-step reduction equals the system–probe mutual information $I(N; J)$ at the current prior q , and by the Holevo bound [16] this is at most the entropy of the probe output, hence at most $\ln(\text{number of probe outcomes}) = \ln(\dim Z)$:

$$\langle \Delta H \rangle \text{ per } Z\text{-transit} = I(N; J) \leq \ln(\dim Z) = \ln 2 = 0.6931\dots \quad (2)$$

This is exactly the ZS-Q7 statement that the Z-channel capacity is $\leq \ln 2$ and that $\Delta S = \ln 2$ per Z-mediated transition, read at the level of acquired pointer information. The Monte Carlo of §8 confirms (2) is never violated over 60 000 random (prior, strength) draws, is *saturated* by a projective measurement, and — scanning the counterfactual probe dimension d — that the ceiling equals $\ln(d)$ to machine precision (ratio 1.000000 for $d = 2\dots 6$). The appearance of $\ln 2$ is therefore **deductively forced by $\dim(Z) = 2$** , not a fitted number. Status of (2): **DERIVED-CONDITIONAL** (Holevo bound combined with the ZS-Q7 capacity identification).

A clarification surfaced by the rate analysis, important for follow-up work: there are *two* distinct “rates.” Equation (2) is the per-transit *information / entropy-reduction* ceiling, which is bounded by $\ln(\dim Z)$. The *asymptotic exponential* convergence rate of the collapse (T1, T3) is a relative entropy between conditional probe distributions and is *not* bounded by $\ln 2$ in general. The I4b bridge of this programme concerns the former; conflating the two would be an error. The weak-coupling regime $\lambda_2 = 2A/Q \approx 0.0146$ nats/transit (ZS-Q15) operates well below the ceiling (2), so in the Z-Spin coupling the collapse is gradual over many transits rather than a single saturating projection; the explicit mapping of measurement strength to λ_2 is **OPEN** (O-Q16.1).

§7. The Strong-Sufficiency Residual: Non-Epistemic Structural Localization

This section is written in deliberate detail for follow-up researchers. It states precisely what weak sufficiency leaves unanswered, proves (by import + simulation) that the residual is *not* an epistemic gap, localizes it structurally, and registers the open gates with promotion paths.

§7.1 The strong question, stated precisely

Weak sufficiency (§5) establishes that a definite outcome occurs and that the ensemble of outcomes is Born-distributed. The strong question asks something more: given the complete physical specification available before the run, is *which* outcome occurs determined, or reducible to a fuller deterministic description? In quantum-trajectory language, the conditional state evolves by the stochastic master equation driven by the measurement innovation (a Wiener process in the diffusive limit, a counting process in the jump limit). The outcome on a given run is fixed by the realized innovation path. The strong question is therefore: is the innovation path eliminable — for instance, fixed by a hidden variable, by fuller knowledge of the initial state, or by a Z-Spin-internal deterministic substrate?

§7.2 The estimation-stability theorem

The decisive structural input is the *estimation-stability* (or filter-stability) property of non-demolition quantum trajectories, proved by Benoist–Pellegrini [4] and in the discrete setting by Bauer–Benoist–Bernard [3] (see also [7]). Informally: two observers who process the *same* measurement record but start from *different* prior states converge to the same conditional state. Formally, let $\rho^\wedge(t)$ and $\tilde{\rho}^\wedge(t)$ be two solutions of the same QND stochastic master equation driven by the same realized record, with $\rho^\wedge(0) \neq \tilde{\rho}^\wedge(0)$ (the second having full support). Under the standard non-demolition and irreducibility hypotheses, $\|\rho^\wedge(t) - \tilde{\rho}^\wedge(t)\| \rightarrow 0$ almost surely, exponentially fast. The limit is the *same random pure state* for both, namely the eigenstate selected by the record.

The interpretive content is sharp: the selected outcome is a **functional of the record**, not of the prior. The prior state influences only the *statistics* of which records occur (Born), not the outcome that any given record produces. Two observers who disagree completely about the initial quantum state nevertheless agree on the outcome once they share the record.

§7.3 Monte-Carlo confirmation

We confirmed estimation stability directly (§8, Block 3b; `zs_q16_outcome_mc_v1_0.py`). For a binary QND measurement via a dim-2 probe, a single record was generated per run and fed to three filters with priors (i) the true Born prior, (ii) an ignorant uniform prior, and (iii) a deliberately wrong prior [0.05, 0.95]. In **2000 / 2000 = 100%** of runs all three filters converged to the *same* outcome — the one selected by the shared record — with the prior-A and prior-B filters agreeing ($L_1 < 10^{-3}$) after on average 12.8 transits. Independently (Block 3a), fixing the initial state and varying only the noise reproduced the Born frequencies (e.g. [0.6038, 0.3962] against Born [0.6, 0.4]): different records give different outcomes. Together these establish empirically that the per-run randomness lives entirely in the record.

§7.4 The residual is non-epistemic

The combination of §7.2 and §7.3 yields the principal negative result of this paper. The strong-sufficiency residual — the irreducibility of “which outcome” — is **non-epistemic**: it is not a deficit of knowledge about the quantum state that a better description could remove. Were the residual epistemic, an observer with a more accurate prior (in the limit, the true state) would obtain a sharper or different prediction of the individual outcome; estimation stability shows this is false — even the true-state observer is led by the record to the same outcome as the worst-informed observer. Consequently, an entire class of would-be resolutions of the strong outcome problem is **excluded within the measurement-dynamics framework**: namely any resolution that proposes to determine the individual outcome by supplementing or sharpening knowledge of the system’s quantum state. The randomness is intrinsic to the record, i.e. to the measurement interaction itself, not to our ignorance of the system.

§7.5 Structural localization at the $\dim(Z)=2$ innovation

Where, then, does the residual reside in Z-Spin? It is the innovation of the single mediating channel, and that channel is the rank-2 = $\dim(Z) = 2$ Z-seam (ZS-Q7; §4.1). The per-run randomness is therefore localized at the two-dimensional Z-boundary: it is the binary innovation of the Z-Spin-mediated handshake. Two quantitative consequences follow. First, the innovation cannot be compressed below the channel: collapsing a system whose pointer entropy is H_0 requires the record to deliver H_0 nats of information, and by (2) each Z-transit delivers at most $\ln 2$, so the minimum record length is $\geq H_0 / \ln 2$ transits — verified in §8 Block 3c for the matched binary case ($H_0 = \ln 2$ collapses in ≈ 9.8 transits against the floor 1.0; $H_0 = 0.35$ in ≈ 6.7 against floor 0.50). Second, when the outcome space exceeds the channel’s binary resolution (more pointer states than the dim-2 probe distinguishes per transit), discrimination slows sharply through aliasing (Block 3c: a 3-state pointer under a dim-2 probe required ≈ 2014 transits), which is the expected signature of the rank-2 bottleneck and is consistent with the corpus identification of the binary outcome space with $\dim(Z) = 2$.

Thus the strong-sufficiency residual is **OPEN** — it is not closed — but it is no longer a vague gap. It is (i) non-epistemic (§7.4), and (ii) structurally identified with the irreducible binary innovation of the $\dim(Z) = 2$ Z-channel, with a quantitative information floor. This is the “converged-but-OPEN” outcome of the corpus exploration discipline: re-examination does not close the node, and the result itself is the confirmation that the present corpus tools localize but do not eliminate the residual.

§7.6 OPEN gates and promotion paths for follow-up work

Table 7.1. Open gates for the strong-sufficiency programme, with explicit promotion paths.

Gate	Open question / promotion path	Current status
O-Q16.1	Derive the per-transit measurement strength (and hence the collapse timescale) from the Z-coupling $\lambda_2 = 2A/Q$, replacing the illustrative calibration of §6. Path: repeated-interaction model with interaction strength fixed by the ZS-Q1 §3.4 / ZS-Q7 §5.8 mediation rates.	OPEN (illustrative only)
O-Q16.2	Decide whether the absolute-strongest form (§7.7) is closable by a Z-Spin-internal mechanism or is provably NON-CLAIM. Path: analyse whether the i-tetration self-reference admits a single-world realization of the record, or whether single-vs-many-world is undecidable within the corpus.	OPEN / candidate NON-CLAIM
O-Q16.3	Upgrade the §4.2 identification (process-level lift witness $\Rightarrow$ selection of the measurement unraveling) from HYPOTHESIS-strong to a rigorous statement relating the ZS-Q14 comb $Y_{\{2:0\}}$ to the choice of unraveling. Path: process-tensor / quantum-comb formalism for the monitored Z-channel.	HYPOTHESIS-strong
O-Q16.4	Extend from the QND (σ_z , non-demolition) case to non-QND measurements where H does not commute with the pointer; characterize the spike/jump statistics (cf. [10]) of the Z-innovation in that regime.	OPEN
O-Q16.5	Identify, within Z-Spin, whether any physical configuration realizes a dark subspace (which would locally defeat purification), or prove that $L_{XY} \equiv 0$ forbids it universally. Path: search for unobserved cross-sector leak channels.	OPEN

What would *falsify* the non-epistemic claim of §7.4: a demonstration that some accessible refinement of the pre-measurement description (a hidden variable, or a sharper state) systematically predicts the individual outcome better than the record alone — i.e. a violation of estimation stability for the physical Z-channel. Within standard quantum mechanics this is excluded by [3, 4]; a Z-Spin-specific violation would require the Z-channel to fail the non-demolition or irreducibility hypotheses, which §4.1 argues it does not. This is registered as gate F-Q16.5.

§7.7 The absolute-strongest form, and the consciousness firewall

Even granting §7.1–§7.6, one question remains untouched: why a definite record exists *at all* — whether the world is single-outcome (one record realized) or branching (all records realized, many-worlds). No measurement dynamics, Z-Spin included, adjudicates this; it is interpretation-dependent. It is retained as **NON-CLAIM** (O-Q16.2 registers the question of whether the corpus can ever decide it). We emphasize, with the corpus discipline, that none of the above is a claim about phenomenal experience. The record, the innovation, and the outcome are operational and channel-structural. The corpus consciousness non-claims NC-Q7.4, NC-A7.6, NC-F10.3, NC-F11.1 and NC-F19.1, and the anti-overclaim gates F-F11.7' (ZS-F11) and F-F19.6 (ZS-F19), remain in force and are not modified by this paper; in particular the ZS-M11 §H16 speculation identifying z^* with self-observation is *not* promoted. The self-referential closure of Q12.A is used here only as the operational generator of the measurement nonlinearity.

§8. Verification Suite

The companion script `zs_q16_outcome_mc_v1_0.py` (numpy only; canonical seed 20260604) implements the repeated-indirect-QND Bayesian filter of [2, 3] — the rigorous model the imported theorems concern — applied to a dim-2 probe. It tests the generic $\dim(Z) = 2$ structure onto which the Z-Spin dictionary (§5) maps; the Z-Spin scalar-tensor dynamics is imported through that dictionary, not re-derived. Results (27/27 PASS across Categories A–E):

Table 8.1. Verification ledger. Category E item (3d) is a control demonstrating the purification condition; the 3-state aliasing of Block 3c is reported as a structural finding, not a failure.

Cat	Checks (key numerical outcome)	Result
A (8)	Locked-input cross-checks: A, Q, (Z,X,Y), $\dim Z$, $w_Y = 6/11$, $\lambda_z = 2A/Q$, $ f'(z^*) = 0.892$, $\ln 2 = \ln(\dim Z)$.	8/8 PASS
B (3)	Weak sufficiency (Block 1): collapse purity 1.000000; fraction $> 0.999 = 1.0000$; Born [0.7035, 0.2965] vs [0.70, 0.30], max dev 0.0035.	3/3 PASS
C (8)	Rate ceiling (Block 2): no violation of $\ln 2$ over 60 000 draws (sup 0.69016); ceiling/ $\ln(d) = 1.000000$ for $d = 2 \dots 6$; among 9 candidate constants only $\ln(\dim Z)$ matches ($\Delta = 0$).	8/8 PASS
D (4)	Strong sufficiency (Blocks 3a–b): randomness-from-record Born [0.6038, 0.3962]; estimation stability 2000/2000 prior-agreement; filter agreement in ≈ 12.8 transits.	4/4 PASS
E (4)	Innovation floor + purification (Blocks 3c–d): $H_0 = \ln 2 \rightarrow 9.8$ transits (floor 1.0); $H_0 = 0.35 \rightarrow 6.7$ (floor 0.5); dark-pair residual 0.6633 (purification fails on dark subspace, as predicted).	4/4 PASS

§9. Anti-Numerology Audit (Protocol 3.1)

No quantity in this paper is a free or fitted parameter. Every number traces to $A = 35/437$, $Q = 11$ and $(Z, X, Y) = (2, 3, 6)$: the outcome-space size is $\dim(Z) = 2$; the rate ceiling is $\ln(\dim Z)$; the Born weight is $\dim(Y)/Q = 6/11$; the Lindblad coefficient is $2A/Q$; the attractor is z^* (ZS-M1). The measurement-strength dependence is *not* fitted — it is left explicitly OPEN (O-Q16.1). A self-correction relative to the originating exploration is recorded here: the rate-ceiling claim was initially tagged “HYPOTHESIS-weak, anti-numerology pending.” The Monte-Carlo design revealed that the ceiling equals $\ln(\dim Z)$ as a *theorem* (the Holevo bound combined with the ZS-Q7 capacity), with the dimensional-locking ratio 1.000000 across $d = 2 \dots 6$ and unique selection among candidate constants. A theorem value has no coincidence to reject; the correct promotion route is the theorem, giving **DERIVED-CONDITIONAL**, and the anti-numerology Monte Carlo is therefore confirmatory of exactness and dimensional specificity rather than a p-value gate. This reclassification is exactly the kind of status-tag correction the corpus auditor (ZS-T11) is designed to surface.

§10. Cross-Paper Consistency and Version-Conflict Check (Protocol 3.2)

Dependency chain audited: ZS-M1 (z^*) $\rightarrow$ ZS-Q12 v4.0 (Q12.A self-referential closure) $\rightarrow$ this paper. ZS-Q16 *imports* z^* and does not modify it; the M1 $\rightarrow$ Q12 $\rightarrow$ Q16 chain is therefore non-circular and conflict-free. Nine cross-paper checks pass (9/9): (1) $L_{XY} \equiv 0$ used as in ZS-F1; (2) rank-2 / capacity $\leq \ln 2$ as in ZS-Q7 Thm 2; (3) Born weight 6/11 as in ZS-Q1 §4.2; (4) σ_z seam-parity pointer as in ZS-Q1 §2; (5) $\chi_Z = -1$ and $\tilde{u}_Z \neq f(\Lambda)$ as in ZS-Q14 Lem 5.1 / §7; (6) $\lambda_2 = 2A/Q$ as in ZS-Q15; (7) NC-F19.7 status refined, not contradicted (ZS-F19 §6.4); (8) the consciousness firewall NC/gates preserved verbatim (ZS-F11, ZS-F19); (9) no new free parameter, A , Q , $\dim(Z)$ LOCKED unchanged. No downstream paper requires modification.

§11. Observational Consistency (Protocol 3.3)

ZS-Q16 makes no new numerical prediction that could conflict with data; at the level of measurement statistics it *reproduces* standard quantum mechanics (Born rule, collapse to eigenstates, CPTP channel), which is consistent with all quantum-measurement experiments. The Penrose–Diósi-type gravitational-decoherence prediction of the corpus ($\tau_D / \tau_{\text{Penrose}} = 1/A = 12.49$; ZS-Q1 §5) is unchanged and remains the testable handle (gold-nanosphere interferometry, 2028–2032). Nothing in this paper bears on Planck 2018 Λ CDM parameters or the Standard-Model couplings, which are fixed in other themes; consistency with them is inherited and not re-litigated here.

§12. Falsification Gates (Protocol 3.5, multi-layered)

Table 12.1. Pre-registered falsification gates, layered as mathematical $\rightarrow$ simulation $\rightarrow$ theoretical, plus a scope/retraction guard.

Gate	Falsification condition	Layer / status
F-Q16.1	The per-Z-transit information gain exceeds $\ln(\dim Z) = \ln 2$ in a correctly specified simulation.	Mathematical (immediate). PASSING: never violated (§8 Cat C).
F-Q16.2	A QND trajectory on the rank-2 Z-channel fails to collapse to a single eigenstate, or the limit statistics depart from Born ($w_Y = 6/11$).	Simulation. PASSING (§8 Cat B, D).

Gate	Falsification condition	Layer / status
F-Q16.3	The process-level witness $\tilde{u}_Z$ is shown to be a function of Λ alone ($\tilde{u}_Z = f(\Lambda)$), removing the basis for record-keeping unraveling selection.	Theoretical. PASSING (ZS-Q14 Cat E; tied to O-Q16.3).
F-Q16.4	A dark subspace is exhibited for the physical cross-sector Z-channel (unobserved leak), defeating purification.	Theoretical / simulation. OPEN guard (O-Q16.5).
F-Q16.5	Estimation stability is violated for the Z-channel: a sharper pre-measurement description predicts individual outcomes better than the record (falsifying the non-epistemic claim of §7.4).	Theoretical. PASSING under [3,4]; anti-overclaim guard for §7.
F-Q16.6	Any promotion of the operational results of §7 to a phenomenal-consciousness claim, or promotion of z^* to self-observation, without new derivation.	Scope / retraction gate (inherits F-F11.7', F-F19.6).

§13. Non-Claims

NC-Q16.1. This paper does not close the strong outcome problem: the per-run outcome is not derived as a deterministic consequence of pre-measurement data; it is shown to be a non-epistemic functional of the record (§7).

NC-Q16.2. This paper makes no claim about the single-world-vs.-many-worlds question (§7.7); the existence of a definite record at all is interpretation-dependent.

NC-Q16.3. No phenomenal-consciousness claim is made. All objects are operational. NC-Q7.4, NC-A7.6, NC-F10.3, NC-F11.1, NC-F19.1 are inherited verbatim; the ZS-M11 §H16 z^* -as-self-observation speculation is not promoted.

NC-Q16.4. Zero new free parameters; the measurement-strength / collapse-timescale mapping to $\lambda_2 = 2A/Q$ is illustrative and OPEN (O-Q16.1), not a fitted constant.

NC-Q16.5. The §4.2 lift-witness $\Rightarrow$ unraveling-selection step is a structural correspondence between two distinct non-uniqueness results, carried at HYPOTHESIS-strong (O-Q16.3); it is not claimed as a theorem.

§14. Conclusion

The single-outcome problem, registered as NC-F19.7, is not one question but three. The Z-Spin geometry closes the first: weak sufficiency — almost-sure collapse to a single eigenstate with Born statistics — is **DERIVED-CONDITIONAL**, because the $L_{XY} \equiv 0$ rank-2 bottleneck makes the unique mediating channel a record-keeping (Boundary-Spin-Lift-selected) QND measurement with no dark subspace, and the rigorous non-demolition convergence theorems then apply, with the i-tetration fixed point z^* as the attractor and a collapse rate ceilinged at the theorem-fixed value $\ln(\dim Z) = \ln 2$. The Z-Spin geometry **localizes but does not eliminate** the second: strong sufficiency is **OPEN**, and — this is the paper's principal structural result — the residual randomness is **non-epistemic**, proved via estimation stability (no refinement of state knowledge alters the per-run outcome), and identified with the irreducible binary innovation of the $\dim(Z) = 2$ Z-seam. The third question — why a definite record exists at all — is bounded out as **NON-CLAIM**. For follow-up researchers, §7.6 records five promotion gates; the most consequential, O-Q16.1, would fix the collapse timescale from $\lambda_2 = 2A/Q$, and O-Q16.2 would decide

whether the corpus can ever speak to the single-world question. No new free parameter was introduced; the consciousness firewall is intact.

§15. How the Universe Records Reality: Four Principles [v1.1 increment]

This section, new in v1.1, consolidates a five-step deep exploration of the question left implicit by §7: if the per-run outcome is recorded by the Z-channel, in what form does the realized world persist? Four principles are registered with their epistemic status. The unifying picture is that the universe records the past not as a stored archive but as a **lossy self-referential cohomological compression**: the realized history is compressed into the present gauge-invariant boundary state, updated cycle-by-cycle by the contracting (leaky) Wilson loop. No v1.0 result is modified.

§15.1 Theorem Q16.R1 — Record Compression

Statement. The realized history of the Z-mediated measurement process is summarized, without loss of any future-relevant information, by the present conditional Z-record state (the filter); the raw observation history is not separately required. Formally, the conditional law π_t of the X-pointer given the Z-record up to time t is a *sufficient statistic* for the future, and the filtering process (π_t) is Markov. Status:

DERIVED-CONDITIONAL.

Grounding. This is the Z-Spin reading of classical nonlinear filtering: the conditional distribution $\pi_t = E[\cdot | F_t^{\{Z\text{-record}\}}]$ is the central object of the Zakai (1969) and Kushner–Stratonovich equations, and by the Kallianpur–Striebel formula the unnormalized filter q_t obeys a *linear* stochastic PDE (the Zakai equation) driven only by the current record increment. The filter is a sufficient statistic and a Markov process [Bain–Crisan 2009], so the present π_t carries all of the past that bears on the future. In Z-Spin the “current conditional Z-record state” of the user’s formulation is exactly this π_t on the $\dim(Z) = 2$ channel; §7’s estimation stability (Q16) is the corresponding statement that the filter forgets its initial condition. Conditional on the identification of the Z-channel innovation with the filtering observation process.

§15.2 Proposition Q16.R2 — Cobordism Ledger

Statement. The physical record is not the gauge-dependent raw boundary wavefunction but the *cohomology class* of the boundary state under the BV-BFV modified quantum master equation — equivalently, the content carried by the cobordism-history fiber $\mathcal{H}_{\text{cob}}$. Status: **HYPOTHESIS-strong**.

Grounding and a dissolved dichotomy. In the BV-BFV formalism the physical boundary state ψ_Σ is *itself* a cohomology class: it satisfies the modified quantum master equation $(\hbar^2 \Delta_\Sigma + \Omega_\Sigma \partial \Sigma) \psi_\Sigma = 0$ [Cattaneo–Mnev–Reshetikhin 2018; Cattaneo–Moshayedi–Wernli 2018, corpus IMPORTED-4], i.e. it is closed under the combined bulk BV Laplacian and boundary BFV operator and defined modulo exact terms. Hence the dichotomy “boundary state vs. cohomology class” dissolves: the gauge-invariant record IS the class $[\psi_\Sigma]$. The Z-Spin cobordism-history fiber $\mathcal{H}_{\text{cob}}$ (ZS-F0 §8.5; ZS-M26 §3, $\mathcal{H}_{\{K,Z\}} = \oplus \chi \mathcal{H}_{\text{cob}} \otimes \mathcal{H}_\chi$) is the corpus carrier of this class. The reason R2 is a Proposition (HYPOTHESIS-strong) and not a Theorem is twofold and is registered as OPEN: (i) the identification of the filter sufficient statistic of R1 with the mQME cohomology class is not yet proved (O-Q16.6); (ii) the Z-Spin Wilson loop is non-unitary (a contraction, §15.4), so the relevant object is a *dissipative* cohomology, which is non-standard and not yet constructed (O-Q16.7). R2 refines, rather than contradicts, the §7/v1.0 picture: the realized X-configuration is the record only through its gauge-invariant cohomological content.

§15.3 Principle Q16.R3 — X-Single / Y-Many Separation

Statement. The realized X-sector carries a single definite record per Z-transit, while the Y-sector carries the unselected possibilities. Status: operational reading **DERIVED-CONDITIONAL**; ontological many-worlds reading **NON-CLAIM**.

The operational separation is a consequence of §5–§7: each Z-transit collapses the X-pointer to one σ_z eigenstate (the recorded outcome), and the unselected branch weight is carried away in the traced-out Y-environment; this is consistent with the corpus sector ontology in which X is the collapsed (particle–space) sector and Y the superposed (wave–time) sector. The *ontological* claim that every Y-branch is an equally real parallel world (the many-worlds reading the user associates with the string-landscape possibility space) is *not* asserted: it is exactly the single-vs.-many question registered as O-Q16.2 and reaffirmed here as NC-Q16.6. R3 is therefore a structural-asymmetry principle, not a multiverse claim, and in particular this paper makes no claim identifying the Y-sector with any specific vacuum count.

§15.4 Principle Q16.R4 — No-Archive Reality

Statement. The universe does not store the past as a growing archive; it compresses the past into the present boundary condition, and the compression is *lossy*. Status: **DERIVED-CONDITIONAL**.

Grounding. Three corpus facts combine. (a) ZS-Q7 §6.2 (Past Hypothesis Unnecessary): the present relaxation toward $p_{eq} = (3,2,6)/11$ is independent of the initial condition, so no stored low-entropy past is needed. (b) R1: the present filter is a sufficient statistic, so the raw history is redundant. (c) The Wilson-loop contraction (ZS-F0 Thm 8.9): the closed $X \rightarrow Z \rightarrow Y \rightarrow Z \rightarrow X$ cobordism has per-cycle dominant-mode survival $|\lambda|^2 = (\pi^2/4)\eta_{topo} = 0.7948 < 1$ with $|\lambda| = |f'(z^*)| = 0.892$, an *attracting* fixed point. **[Erratum E-Q16.1, v1.2: this contraction must NOT be read as global information loss. By ZS-F0 Thm 12.1–12.3 the Wilson loop is a partition function, not a time evolution; $|\lambda|^2 = 0.7948$ is a survival probability, and the complementary $1 - |\lambda|^2 = 0.2052$ is *transferred to the J_Z-odd register mode (slot 1)* — **sum rule $0.7948 + 0.2050 + 0.0001 \approx 1$ — not destroyed. The actual time evolution $U(t) = e^{\{-iHt\}}$ is a separate unitary object (ZS-F0 Thm 12.2, 12.4), so global information is conserved.]****

The record-level forgetting that makes R1/R4 hold is therefore NOT this fine-grained transfer but a distinct *coarse-grained* entropy production (ZS-Q7, $\Delta S = \ln 2$ per Z-transit; NC-Q7.1: a coarse-grained bias, microscopically reversible), under which the global state stays unitary while the X-record loses access to detail. The consistency of these two readings is established in §16. “Records accumulate in the 4D cobordism” is correct only in the weak sense that the cobordism is the kinematic stage (Z-sector) on which the compression runs; it is not an additive store.

§15.5 Synthesis and Reconciliation

Table 15.1. The four recording principles. R1 and R2 are the statistical and topological descriptions of the same object; R4 makes the compression lossy; R3 fixes the X/Y asymmetry while bounding out the multiverse reading.

Principle	One-line content	Status	Key support
R1 Compression	Present conditional Z-record state is a sufficient statistic; raw history redundant.	DERIVED-COND.	Zakai/KS; Q16
R2 Ledger	Record = gauge-invariant cohomology class of the boundary state ($\mathcal{H}_{cob}$).	HYPOTHESIS-stro ng	mQME; M26; O-Q16.6/7

Principle	One-line content	Status	Key support
R3 X/Y	X single record / Y unselected possibilities (operational); many-worlds not claimed.	DERIVED-COND. / NON-CLAIM	Q16 §5–7; O-Q16.2
R4 No-Archive	Past compressed (lossily) into present boundary condition, not stored.	DERIVED-COND.	Q7 §6.2; F0 Thm 8.9

Unified statement. The universe records reality as a **lossy self-referential cohomological compression**: the realized history is compressed (R1, R4) into the gauge-invariant cohomology class of the present X-boundary (R2), the unselected possibilities remaining in Y (R3), with the contracting i-tetration Wilson loop (R4) implementing the per-cycle forgetting. This reconciles the v1.0/§7 reading (“the present configuration is the record”) by identifying the record precisely as the boundary cohomology class rather than the raw configuration.

§15.6 New Gates, Non-Claim, and OPEN Problems (v1.1)

Table 15.2. v1.1 additions: two falsification gates, one non-claim, two OPEN gates.

ID	Statement	Status
F-Q16.7	Falsification of R1/R4: a Z-channel observable is exhibited whose future statistics depend on the raw record history beyond the current conditional state (filter not a sufficient statistic).	PASSING under filtering theory (Zakai/KS).
F-Q16.8	Falsification of R4 lossiness: a correct treatment yields Wilson-loop survival $ \lambda ^2 = 1$ (unitary, lossless), making the record a lossless archive.	PASSING: F0 Thm 8.9 gives $ \lambda ^2 = 0.7948 < 1$.
NC-Q16.6	No ontological many-worlds claim. R3 asserts only the operational X-single / Y-many asymmetry; the reality status of unselected Y-branches is out of scope (inherits O-Q16.2). No identification of Y with any vacuum count is made.	NON-CLAIM (active).
O-Q16.6	Prove the identification $R1 = R2$: that the filter sufficient statistic (conditional Z-record state) equals the BV-BFV mQME cohomology class $[\psi_\Sigma]$. Path: identify the filter state space with the BFV cohomology of the boundary complex.	OPEN (promotes R2 to DERIVED).
O-Q16.7	Construct a dissipative cohomology for the contracting (non-unitary, $ \lambda < 1$) Wilson loop, making R2 well-defined under lossy compression. Path: cohomology of the i-tetration complex with the contraction descending to classes.	OPEN.

§16. Fine-Grained Transfer vs. Coarse-Grained Forgetting: A Consistency Theorem [v1.2 increment]

This section, new in v1.2, resolves the apparent tension flagged by Erratum E-Q16.1 (§15.4): if Z-Spin conserves information, in what sense does the per-cycle Wilson contraction “lose” 20.5%? The resolution, consolidated from a five-step deep exploration, is that the fine-grained (global, unitary) and coarse-grained (record-level) descriptions are the same physics at two resolutions, and that the 20.5% is transferred within the register, not deposited in the Y-sector.

§16.1 Destination of the per-cycle 20.5%

By ZS-F0 Thm 12.3 the per-cycle amplitude budget decomposes as $|\lambda|^2 + (J_Z\text{-odd residual}) + (X\text{--}Y \text{ intra-block leak}) = 0.7948 + 0.2050 + 0.0001 \approx 1$. The complementary $1 - |\lambda|^2 = 0.2052$ is **transferred to the J_Z-odd register mode (slot 1)**, which lies inside the Z-block, not to the Y-sector; the Hodge reading (ZS-M27 §6.2, HYPOTHESIS-strong) identifies it with the BRST-exact image $\text{im } D$ (the gauge / ghost orbit). The only Y-touching term is the X–Y intra-block residual 0.0001, suppressed by $L_{XY} \equiv 0$. Therefore the natural intuition that the lost amplitude is “fed back into the Y possibility space” is *not* what the corpus states (NC-Q16.7). What is true is a topological fact about the closed loop: the Wilson cobordism $X \rightarrow Z \rightarrow Y \rightarrow Z \rightarrow X$ *re-traverses* Y every cycle, but that Y-traversal is internal to the surviving 0.7948 eigenmode of the whole loop — it is not an additional deposit of the 20.5% into Y.

§16.2 Theorem Q16.C1 — Two-Resolution Consistency

Statement. The fine-grained description (global evolution $U(t) = e^{\{-iHt\}}$ unitary, partition-function survival $|\lambda|^2 = 0.7948$, complementary 0.2052 transferred to the J_Z-odd mode; ZS-F0 Thm 12.1–12.4, PROVEN) and the coarse-grained description (the X-record is a sufficient statistic that forgets detail at rate $\Delta S = \ln 2$ per Z-transit; ZS-Q7, R1/R4) are **mutually consistent**: they are one physical process viewed at two resolutions, related by the Stinespring dilation / GKSL reduction in which the J_Z-odd (ghost) mode plays the role of the traced-out environment. Status: **DERIVED-CONDITIONAL**.

Grounding. The corpus already realizes the Z-mediated channel as a Stinespring dilation $\Phi_Z(\rho) = \text{Tr}_Z[V(\rho \otimes |0\rangle\langle 0|)V^\dagger]$ (ZS-Q1 §3.3; ZS-Q7 §8), with V a global isometry. Two imported PROVEN facts close the consistency. (i) Stinespring (1955): tracing the ancilla of a global unitary / isometry yields a CPTP channel; the global state stays pure while the reduced state need not. (ii) GKSL / Lindblad (1976): the reduced dynamics of a subsystem is a completely-positive trace-preserving semigroup, irreversible, valid on time scales below the recurrence time of the closed system $S + R$, even though $S + R$ evolves unitarily. Reinterpreting the X-record as the “system” and the J_Z-odd ghost mode (plus the Y bulk) as the “environment”: the fine-grained transfer of 0.2052 into the ghost mode is unitary and reversible up to recurrence, but the X-record cannot resolve the ghost mode, so the same transfer is read by the record as coarse-grained entropy production $\Delta S = \ln 2$ (ZS-Q7 NC-Q7.1: a coarse-grained bias, microscopically reversible). Hence the fine-grained von Neumann entropy is conserved while the coarse-grained record entropy increases — no contradiction. Conditional on the identification of the traced-out environment with the J_Z-odd / Y bulk modes.

§16.3 What is NOT Identified

The per-cycle amplitude fraction $1 - |\lambda|^2 = 0.2052$ (dimensionless) and the per-transit entropy production $\ln 2 \approx 0.693$ nats are **different quantities at different layers** and are not claimed equal; the consistency of §16.2 is structural (same process, two resolutions), not the numerical identity $0.2052 = \ln 2$ (which is false). The 20.5% is a fine-grained register-mode transfer; the $\ln 2$ is a coarse-grained information-theoretic production. Conflating the two was the imprecision corrected by Erratum E-Q16.1.

Table 16.1. The two readings are one process at two resolutions; the fine-grained transfer is the coarse-grained forgetting seen through the partial trace over the J_Z-odd / Y modes.

Resolution	Object	Information behaviour
Fine-grained	Global pure state on $H_X \otimes H_Z \otimes H_Y$; $U(t) = e^{\{-iHt\}}$ unitary; Wilson partition function.	Conserved. $S_{vN} = \text{const. } 0.2052$ transferred to J_Z-odd mode (sum rule ≈ 1). Reversible up to recurrence.

Resolution	Object	Information behaviour
Coarse-grained	X-record sufficient statistic (filter); ghost + Y bulk traced out (Stinespring / GKSL).	Record forgets detail. S_{coarse} rises by $\Delta S = \ln 2$ per Z-transit (Q7). Irreversible at record level.

§16.4 New Erratum, Non-Claim, and Gate (v1.2)

Table 16.2. v1.2 additions: one erratum, one non-claim, one falsification gate.

ID	Statement	Status
E-Q16.1	Erratum (v1.2). The §15.4 phrase that ~20% of boundary information is “forgotten” per cycle is corrected: $ \lambda ^2 = 0.7948$ is a partition-function survival probability, the complementary 0.2052 is transferred to the $J_{\text{Z-odd}}$ mode (ZS-F0 Thm 12.1–12.3), and the global time evolution $U(t)$ is unitary (Thm 12.2, 12.4). Record-level forgetting (R1/R4) is the distinct coarse-grained entropy production of ZS-Q7.	ERRATUM (documented).
NC-Q16.7	No global information loss, and no Y-deposit. The per-cycle Wilson contraction is information transfer (to the $J_{\text{Z-odd}}$ mode) plus coarse-grained forgetting, not a violation of global unitarity; the 20.5% is not fed into the Y possibility space (only the 0.0001 X–Y residual touches Y). Cites ZS-F0 Thm 12.1–12.4, ZS-Q7 NC-Q7.1.	NON-CLAIM (active).
F-Q16.9	Falsification of Theorem Q16.C1: a correct treatment shows the per-cycle 0.2052 is irreversibly destroyed (no register-mode transfer, sum rule < 1) OR is deposited in the Y-sector rather than the $J_{\text{Z-odd}}$ mode.	PASSING: F0 Thm 12.3 sum rule ≈ 1 to $J_{\text{Z-odd}}$ mode.

§17. The Z_2 -odd Mode: Ghost–Environment Identification, R2 Promotion, and the Backbone Reading [v2.0 increment]

This section, new in v2.0, consolidates a five-step deep exploration of the $J_{\text{Z-odd}}$ register slot (slot 1). It (i) discharges the conditional of Theorem Q16.C1, promoting it to **DERIVED**; (ii) grounds the cohomology framing of Proposition Q16.R2, promoting it to **DERIVED-CONDITIONAL**; and (iii) registers the “Backbone” reading of the Z_2 -odd mode with explicit status. No prior content is modified.

§17.1 The Z_2 -odd slot, concretely

Slot 1 is the $J_{\text{Z}} = -1$ eigenspace of $J_{\text{Z}} = \text{diag}(+1, -1, +1, \dots)$ (ZS-F0 §8.6), the register image of the $\varepsilon \rightarrow -\varepsilon$ projection (ZS-S1 §5.2), and the Bargmann–Fock first-excited mode $|1\rangle_{\text{F}}$ of the Z-block (ZS-F0 §9.1). Three corpus-PROVEN facts pin down its role. (a) **BRST-exact / ghost**: the per-cycle leak $P_{\text{phys}} = |1\rangle\langle\ell_{\text{W}}|$, with $\langle\ell_{\text{W}}| \approx 0.6885 \langle 0_{\text{Z}}|$ matching $\text{Im}(\lambda) = (\pi/2)\text{Re}(z^*)$, is BRST-exact on the register+ghost complex with the ghost pair $|1\rangle\langle b|$ and $Q_{\text{0}}^2 = 0$ (ZS-M22 §6.6.4, rank-one closure PASS; Hodge weight $\text{im } D$, ZS-M27 §6.2). (b) **X/Y-decoupled**: in the explicit 11×11 Wilson matrix the slot-1 Z_2 -odd component does not couple to the X–Y cross-block (ZS-F0 §8.8). (c) **observable-invisible**: the $J_{\text{Z-odd}}$ content is annihilated by the Z-mediator sandwich $\Pi_{\text{Z}}(\cdot)\Pi_{\text{Z}}$ and is invisible to the corpus bilinear observable form (ZS-M31.4, DERIVED). Together: slot 1 carries gauge / ghost content that neither sources X/Y nor registers in observables.

§17.2 Path A — Ghost = Traced-Out Environment (Theorem Q16.C1 $\rightarrow$ DERIVED)

Theorem Q16.C1 (§16.2) was DERIVED-CONDITIONAL on identifying the traced-out environment of the Stinespring / GKSL reduction with the J_Z-odd (ghost) mode. The Stinespring dilation $\Phi_Z(\rho) = \text{Tr}_Z[V(\rho \otimes |0\rangle\langle 0|)V^\dagger]$ (ZS-Q1 §3.3) traces over the Z-sector; by §17.1 the surviving Z-content is the J_Z-even mode (slot 0, $|v_W\rangle$ direction, harmonic 0.7948) and the discarded content is the J_Z-odd mode (slot 1, BRST-exact 0.2052). By the imported PROVEN physical-state construction (Kugo–Ojima 1979; Henneaux–Teitelboim 1988/1992), BRST-exact states are null / unphysical — precisely “environmental” content that is traced away and gauged out of the physical record. The conditional is therefore discharged: the ghost mode (slot 1) *is* the unphysical part of the traced-out environment that absorbs the per-cycle transfer and is invisible to the X-record (§17.1c). Status: Theorem Q16.C1 promoted DERIVED-CONDITIONAL → **DERIVED**.

§17.3 Path B — Filter = BRST Cohomology (Proposition Q16.R2 → DERIVED-CONDITIONAL)

The physical record is the BRST cohomology $H^0(Q_{\text{BRST}}) = \ker Q_B / \text{im } Q_B$ (Kugo–Ojima; Henneaux–Teitelboim, PROVEN), which is the J_Z-even / harmonic content (0.7948) with the ghost ($\text{im } D$, 0.2050) excluded. The filter sufficient statistic of R1 is gauge-invariant and does not track the J_Z-odd content (ZS-M31.4). Hence R1 (statistical record) and R2 (cohomological record) **agree on the excluded set** — both quotient out exactly the BRST-exact ghost. This is a genuine partial bridge: “record = cohomology class” is no longer a bare hypothesis but the Kugo–Ojima physical-state quotient, and R1’s exclusion of the ghost matches $\text{im } Q_B$ exactly. What remains OPEN is the full functorial bijection between the filter state space (probability/density operators on the signal) and $H^0(Q_{\text{BRST}})$; this residual is registered as O-Q16.6’ (subsuming O-Q16.6, and joined by the dissipative-cohomology O-Q16.7). Status: Proposition Q16.R2 promoted HYPOTHESIS-strong → **DERIVED-CONDITIONAL**.

§17.4 The Backbone Reading: What Holds and What Does Not

The proposed reading of slot 1 as a “Backbone” splits cleanly under the analysis above. Two parts hold; one must be qualified.

Table 17.1. The Backbone reading: observational invisibility and ontological preservation are DERIVED; the “accumulating archive” extrapolation is a NON-CLAIM (NC-Q16.8).

Backbone claim	Verdict	Status
(1) Observationally forgotten	Slot 1 does not output to X or Y and is invisible to observables (ZS-F0 §8.8; ZS-M31.4; BRST-exact $\Rightarrow$ null norm, Kugo–Ojima). From the X/Y observer it is effectively forgotten.	DERIVED
(2) Ontologically preserved	Not removed from the universe: global $U(t) = e^{-iHt}$ is unitary and the sum rule ≈ 1 conserves the amplitude (ZS-F0 Thm 12.2–12.4).	DERIVED
(3) Accumulates as topological backbone	As a structural channel of the Z-sector 2D stage (the $J_Z = -1$ / Bargmann–Fock $ 1\rangle_F$ mode) the role is defensible. But “accumulates as a growing archive” conflicts with R4 (No-Archive) and with BRST gauge-redundancy (an exact mode carries no physical, accumulable information); it must be	structural role HYP-strong: “growing archive” NON-CLAIM

Backbone claim	Verdict	Status
	read as a steady-state gauge channel, not a store. Partial identification with the Z-sector holonomy invariant $\eta_{\text{topo}} = \ \mathbb{z}^*\ ^2$ is interpretive.	

§17.5 Status Upgrades, New Non-Claim, and OPEN Problems (v2.0)

Table 17.2. v2.0: two status upgrades (C1, R2), one non-claim, two OPEN problems.

ID	Statement	Status
C1 ↑	Theorem Q16.C1 (fine/coarse consistency) is promoted: its environment-identification condition is discharged by the slot structure (§17.1–17.2).	DERIVED-CONDITIONAL → DERIVED
R2 ↑	Proposition Q16.R2 (Cobordism Ledger) is promoted: “record = BRST cohomology class” is grounded in Kugo–Ojima / Henneaux–Teitelboim, with R1/R2 agreeing on the ghost exclusion (§17.3).	HYPOTHESIS-strong → DERIVED-CONDITIONAL
NC-Q16.8	The $\mathbb{Z}_2$ -odd (ghost) mode is NOT a growing physical archive. It is a gauge-redundant (BRST-exact) steady-state channel of the Z-sector; the “accumulation” reading conflicts with R4 (No-Archive) and with BRST null-norm gauge-redundancy. Observational invisibility and ontological preservation are the only DERIVED Backbone claims.	NON-CLAIM (active).
O-Q16.6'	Prove the full functorial bijection between the filter state space (R1) and the BRST cohomology $H^0(Q_{\text{BRST}})$ (R2). Subsumes O-Q16.6; promotes R2 to DERIVED if closed. Path: identify the gauge-invariant filter algebra with $\ker Q_{\text{B}} / \text{im } Q_{\text{B}}$ on the cobordism-history fiber.	OPEN.
O-Q16.8	Prove that the per-cycle $\mathbb{J}_{\text{Z-odd}}$ transfer (0.2052) does not monotonically accumulate — i.e. that slot 1 reaches a steady state rather than a growing reservoir — making NC-Q16.8 a theorem. Path: spectral analysis of the $\mathbb{J}_{\text{Z-odd}}$ component under iterated W.	OPEN.

§18. The $\mathbb{J}_{\text{Z-odd}}$ Leak as Attractor Damping: Necessity for Collapse and the Arrow of Time [v2.1 increment]

This section, new in v2.1, resolves a question that the §15–§17 picture leaves sharp: if the per-cycle $\mathbb{J}_{\text{Z-odd}}$ transfer (20.5%) is gauge-redundant, why does the universe route amplitude into it at all — is it waste? The answer, consolidated from a five-step deep exploration, is that the leak is the **damping that makes z^* an attracting fixed point**, hence a necessary condition for collapse and the arrow of time. It is gauge-redundant yet functionally essential — not waste. This is registered as Theorem Q16.C2 and the guarding NC-Q16.9.

§18.1 The leak is the attractor damping

ZS-M1 Remark 1.2 (PROVEN) states that the Wilson loop leaks by $1 - |\lambda|^2 = 1 - (\pi^2/4)\eta_{\text{topo}} \approx 0.2052$ per cycle, and that *this leakage is the damping that makes z^* an attracting fixed point*; a closed loop ($|\lambda|^2 = 1$) gives only marginal stability. By Koenigs (1884) and standard complex dynamics, a holomorphic map has an *attracting* fixed point precisely when its multiplier satisfies $0 < |f'(z^*)| < 1$; $|f'| = 1$ is the neutral (marginal) case with no local convergence. The Z-Spin multiplier is $f'(z^*) = \lambda$ with $|\lambda| = 0.8915 < 1$

(ZS-M1 L5), and the principle scan of ZS-Q12 §5 confirms z^* is strictly sub-critical (emergent criticality $|\lambda| = 1$ is a documented No-Go). The leak $1 - |\lambda|^2 > 0$ is therefore exactly the deficit that places $|\lambda| < 1$ — it is the analytic content of the attracting property.

§18.2 Theorem Q16.C2 — Necessity for collapse and the arrow of time

Statement. Within the corpus identification of wave-function collapse as convergence to the attracting fixed point z^* (ZS-Q12.A; Belavkin), the per-cycle J_Z -odd leak $1 - |\lambda|^2 = 0.2052 > 0$ is a **necessary condition** for collapse: an attracting z^* requires $|\lambda| < 1$ (Koenigs), equivalently a nonzero leak; a non-leaky loop ($|\lambda| = 1$) is marginal and does not converge, so no definite outcome forms. Corollary: the same irreversible convergence is the microscopic face of the structural arrow of time (ZS-Q7, $\Delta S = \ln 2$ per Z -transit). The leak is gauge-redundant (BRST-exact, no observable information; Kugo–Ojima) yet functionally essential. Status: **DERIVED-CONDITIONAL** (on the ZS-Q12.A collapse = z^* -convergence identification, itself conditional on the 4π spin-closure axiom; the arrow corollary is conditional on coarse-graining, ZS-Q7 NC-Q7.1).

Proof sketch. (1) Collapse = convergence to z^* (ZS-Q12.A, 92% basin). (2) Local convergence to a fixed point of a holomorphic self-map requires the multiplier to lie strictly inside the unit disk, $0 < |f'(z^*)| < 1$ (Koenigs 1884); $|f'| = 1$ is neutral with no contraction. (3) For Z -Spin, $f'(z^*) = \lambda$ and $|\lambda|^2 = 0.7948$, so $1 - |\lambda|^2 = 0.2052 > 0$ is precisely the statement $|\lambda| < 1$. (4) Contrapositive: if the leak vanished ($|\lambda| = 1$), z^* would be marginal, convergence would fail, and no collapse could occur. Hence the leak is necessary for collapse. (5) The relaxation is monotone and irreversible at the coarse-grained level (ZS-Q7), so the same leak underwrites the arrow of time.

§18.3 NC-Q16.9 — Gauge-redundant yet functionally essential (not waste)

The leak is BRST-exact (im D; ZS-M27 §6.2, ZS-M22 §6.6.4) and hence null-norm / unphysical (Kugo–Ojima): it carries no independent observable information. NC-Q16.9 makes explicit that this does NOT make it waste. It is not (a) global information loss — it is transferred to the J_Z -odd mode with the sum rule ≈ 1 (v1.2, ZS-F0 Thm 12.3); not (b) an informational reference or pointer — gauge-redundant content cannot be read as a value (v2.0); and not (c) a growing archive — it is a steady-state gauge channel (NC-Q16.8). What it IS: the functional damping channel that (i) makes z^* attracting, hence purchases collapse and the arrow of time (Theorem Q16.C2), (ii) absorbs unphysical modes to preserve unitarity via the Kugo–Ojima quartet, and (iii) houses the Wilson–Berry–Keating non-commutativity, since $[W, L_{\{1/2\}}]_Z$ resides entirely in the J_Z -odd grading (ZS-F0 §8.7, PROVEN). “No observable information” is not “no function” — exactly as a gauge degree of freedom in QED is unobservable yet essential to consistency.

§18.4 The 80/20 Reading and Supporting Functions

Table 18.1. The definite 79.5% output cannot exist without the 20.5% damping; they are two faces of one contraction. The unobservable 20.5% is where definiteness, unitarity, and the arrow of time are generated — production, not waste.

Channel	Role	Status
79.5% (Z_{even})	The observable output: the harmonic mode projected to X , i.e. the definite realized reality.	the visible result
20.5% ($J_Z\text{-odd}$)	The engine: the attractor damping that drives convergence (collapse, Thm Q16.C2), the	the machinery

Channel	Role	Status
	quartet that preserves unitarity, and the locus of the non-commutativity $[W, L_{\{1/2\}}]$ (ZS-F0 §8.7).	

§18.5 New Theorem, Non-Claim, Gate, and OPEN (v2.1)

Table 18.2. v2.1 additions: one theorem (Q16.C2), one non-claim, one falsification gate, one OPEN problem.

ID	Statement	Status
Q16.C2	The per-cycle J_Z -odd leak $1 - \lambda ^2 = 0.2052$ is the z^* -attractor damping (ZS-M1 Rem 1.2; Koenigs 1884) and a necessary condition for collapse (= convergence to z^* , ZS-Q12.A) and the arrow of time (ZS-Q7); gauge-redundant yet functionally essential.	DERIVED-CONDITIONAL
NC-Q16.9	No observable information $\neq$ no function. The J_Z -odd leak is gauge-redundant (BRST-exact, null-norm) yet functionally essential; it is not waste, not global information loss, not an informational reference, and not a growing archive.	NON-CLAIM (active).
F-Q16.10	Falsification of Theorem Q16.C2: wave-function collapse (a definite outcome) is exhibited with $ \lambda = 1$ (no leak, marginal loop), OR the leak is shown to be unnecessary for convergence to z^* .	PASSING: $ \lambda ^2 = 0.7948 < 1$ (ZS-M1); collapse = z^* convergence (ZS-Q12.A).
O-Q16.9	Remove the conditionality of Theorem Q16.C2 by deriving collapse = z^* -convergence (ZS-Q12.A) without assuming the 4π spin-closure axiom — i.e. promote C2 from DERIVED-CONDITIONAL to DERIVED.	OPEN.

§19. Parallel Closure Attempt on O-Q16.9 and O-Q16.6': A Confirmed-OPEN Result [v2.2 increment]

This section, new in v2.2, reports a five-step parallel deep exploration that attempts to close the two outstanding promotion gaps: O-Q16.9 (derive collapse = z^* -convergence without the 4π -closure axiom, promoting Theorem Q16.C2 to DERIVED) and O-Q16.6' (prove the filter sufficient statistic equals the BRST cohomology class, promoting Proposition Q16.R2 to DERIVED). The honest outcome is that **neither closes with current corpus tools**. Per the corpus discipline, a converged exploration whose closure nodes remain OPEN is itself a valid result — a confirmed-genuine-OPEN, i.e. a precise statement of what cannot yet be derived. No promotion of C2 or R2 is made; two substantive clarifications are registered.

§19.1 O-Q16.9 — The conditionality relocates to $Z=\partial X$ but does not vanish

Three corpus results bound the attempt. (a) ZS-Q12 §5 (PROVEN No-Gos): no emergent principle forces λ — emergent criticality requires $|\lambda| = 1$ (but $|\lambda| = 0.8915$, sub-critical), the convergence rate is monotone (no variational extremum), and a continuum of generator bases is attracting (no attractor-uniqueness). So λ , hence collapse = z^* -convergence, cannot be obtained by emergence alone. (b) ZS-M1 Remark 1.3 (PROVEN): the decoherence / CPTP route converges to $z \rightarrow 0$, not z^* — i-tetration is pre-quantum classical geometry — so collapse-to- z^* cannot be replaced by a generic open-system relaxation. (c) ZS-Q12V §14, Theorem q12.bdy (DERIVED): the 4π spin-closure is *not* an independent axiom; it is DERIVED from $Z=\partial X$ together with $(Z, X) = (2, 3)$ via the ambient frame bundle $SO(3)$, $\pi_1(SO(3)) = \mathbb{Z}/2$,

and the $SU(2)$ double cover ($b = i$, since $i^2 = -1$ is the 2π sign-flip and $i^4 = 1$ the 4π return). By ZS-Q12V §16, the single irreducible postulate of the whole programme is $Z=\partial X$ (equivalently: Z separates X from Y , by Jordan–Brouwer; or the X – Y transfer is holographic, by Bekenstein).

Consequence for C2. The conditionality of Theorem Q16.C2 (stated in §18 as resting on the 4π -closure) **relocates**: since 4π is itself DERIVED from $Z=\partial X$, C2’s only residual condition is $Z=\partial X$ — the corpus’s single foundational geometric postulate, shared by every DERIVED result in the framework, not an extra assumption special to C2. This is a genuine tightening, but it is not a promotion: $Z=\partial X$ remains an irreducible postulate, so C2 stays **DERIVED-CONDITIONAL** (now conditional on $Z=\partial X$ rather than on a separate 4π axiom). O-Q16.9 is therefore confirmed OPEN.

§19.2 O-Q16.6’ — The obstruction sharpens to a two-cohomology mismatch but does not close

The partial bridge of v2.0 §17.3 stands: the filter (R1) and the BRST record (R2) agree on the excluded set — both quotient out the J_Z -odd ghost ($\text{im } Q_B$). Two further facts sharpen, but do not close, the gap. (a) The filter independently carries a cohomological structure: by information cohomology (Baudot–Bennequin 2015; Vigneaux 2020), Shannon entropy is the unique degree-1 cohomology class on an information structure, and for continuous observables the degree-1 cocycles are the differential entropy together with the dimension of the measure support. So R1 is cohomological in its own right. (b) However, information cohomology (a derived functor on presheaves over a category of observables) is a **different cohomology theory** from BRST cohomology (H^0 of the gauge-constraint complex $\ker Q_B / \text{im } Q_B$). Identifying the two is the residual obstruction, and it carries a concrete dimensional mismatch: the filter state space is a continuum (probability measures / density operators), whereas $H^0(Q_BRST)$ is finite-dimensional (ZS-M27 §7: $\dim \ker D = 4$, chirality $\dim H_D^{\pm} = 2$). A naive bijection therefore fails. Proposition Q16.R2 stays **DERIVED-CONDITIONAL**; O-Q16.6’ is confirmed OPEN.

§19.3 The two OPENs meet at z^* (new structural observation)

The dimensional mismatch of §19.2 points to where a correspondence could survive: not at the level of full state spaces, but at the **fixed-point / asymptotic** level. The filter’s collapse fixed points — the z^* -convergence of Theorem Q16.C2 — are a finite, discrete set, and could plausibly stand in bijection with the finite-dimensional BRST cohomology classes. If so, O-Q16.6’ (filter = BRST cohomology) and O-Q16.9 (collapse = z^* -convergence) are two faces of one statement, joined through z^* . This is a structural observation, not a proof: it identifies the sharpened next target (the z^* -level filter–cohomology correspondence) registered as O-Q16.10, whose closure would advance both promotions at once. Status: HYPOTHESIS.

§19.4 Status — Confirmed-OPEN, No Fabricated Closure (v2.2)

Table 19.2. v2.2: two OPENs confirmed (no promotion), two conditionality clarifications, one non-claim, one new OPEN problem.

ID	Statement	Status
O-Q16.9	Confirmed OPEN. Collapse = z^* -convergence cannot be derived without a geometric postulate (emergence No-Go, ZS-Q12 §5; decoherence $\rightarrow z \rightarrow 0$, ZS-M1 Rem 1.3). Clarification: the 4π -closure is DERIVED from $Z=\partial X$ (ZS-Q12V §14), so C2’s condition relocates to the single corpus postulate $Z=\partial X$.	CONFIRMED-OPEN.

ID	Statement	Status
O-Q16.6'	Confirmed OPEN. Sharpened: the filter carries an information-cohomology structure (Baudot–Bennequin 2015) distinct from BRST cohomology, with a continuum-vs-finite-dimensional mismatch (ZS-M27 §7). Partial bridge (ghost-exclusion agreement) stands.	CONFIRMED-OPEN.
Q16.C2	Unchanged status; conditionality clarified to rest on $Z=\partial X$ (not a separate 4π axiom). No promotion.	DERIVED-CONDITIONAL (on $Z=\partial X$)
Q16.R2	Unchanged status; obstruction sharpened to the two-cohomology identification. No promotion.	DERIVED-CONDITIONAL
NC-Q16.10	v2.2 does NOT promote Theorem Q16.C2 or Proposition Q16.R2. The two OPENs (O-Q16.9, O-Q16.6') are confirmed genuine — termini with current corpus tools — not closed. Reaching a precise terminus is a result; fabricating a closure is not.	NON-CLAIM (active).
O-Q16.10	Establish the z^* -level correspondence: the filter's collapse fixed points (z^* -convergence, C2) $\leftrightarrow$ the BRST cohomology classes (R2). Closure would simultaneously advance O-Q16.9 and O-Q16.6' through z^* .	OPEN.

§20. The State-on-Cohomology Functor and the z^* -Level Closure [v2.3 increment]

This section, new in v2.3, closes the two problems v2.2 left CONFIRMED-OPEN — O-Q16.6' (filter $\leftrightarrow$ BRST record) and O-Q16.10 (collapse fixed points $\leftrightarrow$ cohomology classes) — by correcting the category. The v2.2 obstruction (§19.2) was a **category error**: it sought a bijection between the filter state space (a continuum of probability measures) and the BRST cohomology $H^0(Q_BRST)$ (a finite-dimensional vector space). The right target is not the cohomology vector space but the **state functor on the BRST physical pointer algebra**. Because the state space of a finite-dimensional commutative C^* -algebra is itself a continuum (a simplex), the continuum-vs-finite-dimensional mismatch dissolves. Theorem Q16.C2 is NOT promoted by this section; it remains **DERIVED-CONDITIONAL** on $Z=\partial X$.

§20.1 The BRST physical pointer algebra (Lemma Q16.F1)

Lemma Q16.F1 (BRST Pointer Algebra). The Z -seam pointer observable generates a commutative algebra $A^0_{\{BRST,Z\}} = \langle P_0, P_1 \rangle \cong \mathbb{C}^2$, where P_0, P_1 are BRST-closed, non-exact, mutually orthogonal primitive idempotents ($P_i P_j = \delta_{ij} P_i$, $P_0 + P_1 = 1$) selected by the Z -seam parity σ_z . Grounding: the pointer is the Kraus-index parity — which of the two Kraus operators $\{K_0, K_1\}$ is occupied (ZS-A7 §3.2-bis SE-3) — with exactly two outcomes because $\dim(Z) = 2$ (ZS-Q1 §3.3 L7); P_0, P_1 project onto the two σ_z eigenstates inside the physical sector $H^0(Q_BRST)$, selecting two of the $\dim H_D = 4$ cohomology classes (ZS-M27 §1, the chirality-split V_4 basis). Status: **DERIVED-CONDITIONAL** (on the identification of the BRST physical sector with the Z -seam pointer basis).

§20.2 Filter as state (Lemma Q16.F2)

Lemma Q16.F2 (Filter-as-State). Every binary QND filter posterior $q_t = (q_0, q_1)$ (ZS-Q16 Appendix A) defines a unique state ω on $A^0_{\{BRST,Z\}}$ by $\omega(P_n) = q_n$, and every state on $A^0_{\{BRST,Z\}} \cong \mathbb{C}^2$ arises uniquely this way. The state space $\text{State}(\mathbb{C}^2) = \Delta_1 = \{(q_0, q_1) : q_n \geq 0, q_0 + q_1 = 1\}$ is a one-parameter continuum — the same continuum the filter posterior lives in. Status: **PROVEN** (the states of a finite-dimensional commutative C^* -algebra $\mathbb{C}^n$ form the $(n-1)$ -simplex; Gelfand–Naimark 1943). This is

exactly the step that dissolves the §19.2 mismatch: the filter is compared with a state space, not a vector space.

§20.3 Bayes update = BRST algebraic conditioning (Lemma Q16.F3)

Lemma Q16.F3 (Conditioning Commuting Square). For each probe outcome j , define the measurement effect $E_j = \sum_n p(j|n) P_n \in A^0\{\text{BRST}, Z\}$. The algebraic (a posteriori) conditioning of the state, $\omega'(A) = \omega(E_j A) / \omega(E_j)$, reproduces the Bayes update of the filter exactly: setting $A = P_n$ gives $\omega'(P_n) = p(j|n) q_n / \sum_m p(j|m) q_m$, the Bayes posterior. Status: **PROVEN** algebraically (commutativity makes E_j and P_n simultaneously diagonal, so the symmetrized a-posteriori state of Ozawa 1985 collapses to Bayes); **DERIVED-CONDITIONAL** for the Z-Spin physical identification of E_j with the Z-mediated probe.

§20.4 Full functorial bijection — closing O-Q16.6' (Theorem Q16.F4)

Theorem Q16.F4 (Full Functorial Bijection). The binary Z-channel filtering process Filt_Z is naturally isomorphic to the state functor on the BRST physical pointer algebra, $\text{Filt}_Z \cong \text{State}(A^0\{\text{BRST}, Z\})$, with update maps preserved (the Bayes square of F3 commutes with conditioning at every step, hence under composition). This is the correct closure of O-Q16.6': the filter sufficient statistic is the BRST physical state, not the cohomology vector. Status: **DERIVED-CONDITIONAL**. Consequence: O-Q16.6' is **CLOSED** (in the correct category), and Proposition Q16.R2 (Cobordism Ledger) is strengthened to **DERIVED-CONDITIONAL** — “record = BRST physical class” now realized as the state/observable duality on $A^0\{\text{BRST}, Z\}$. The naive vector-space bijection is explicitly NOT claimed (NC-Q16.11; falsified by F-Q16.11).

§20.5 z^* -level correspondence — closing O-Q16.10 (Theorem Q16.F5)

Theorem Q16.F5 (z^* -Level Filter–BRST Correspondence). At the asymptotic collapse level, the QND filter converges almost surely to a pointer vertex, $\text{Fix}_{\{z^*\}}(\text{Filt}_Z) = \{\delta_0, \delta_1\}$ (ZS-Q12.A; ZS-Q16 §7). These two endpoints are naturally bijective with the primitive characters of $A^0\{\text{BRST}, Z\}$ — the two evaluation functionals $\chi_n(P_m) = \delta_{nm}$ — equivalently the two Z-seam BRST cohomology classes selected by σ_z : $\delta_n \leftrightarrow \chi_n \leftrightarrow [\psi_n]$. Thus O-Q16.6' and O-Q16.10 are two faces of one statement, joined through z^* . Status: **DERIVED-CONDITIONAL** on $Z=\partial X$ (the z^* -convergence inherits the Theorem Q16.C2 conditionality). Consequence: O-Q16.10 is **CLOSED**. The §19.3 z^* -observation is now a theorem, not a hypothesis.

§20.6 Status, Erratum, Non-Claim, and Gate (v2.3)

Table 20.1. v2.3: two OPENs closed (F4, F5) in the correct category, R2 strengthened, C2 unchanged; one erratum, one non-claim, one gate, one residual OPEN.

ID	Statement	Status
O-Q16.6' ↓	CLOSED via Theorem Q16.F4: the filter sufficient statistic is naturally isomorphic to the state functor on $A^0\{\text{BRST}, Z\} \cong \mathbb{C}^2$, with Bayes = BRST conditioning. Not a vector-space identity.	CONFIRMED-OPEN → CLOSED (F4)
O-Q16.10 ↓	CLOSED via Theorem Q16.F5: the z^* -collapse fixed points $\{\delta_0, \delta_1\}$ are bijective with the primitive characters / Z-seam cohomology classes.	OPEN → CLOSED (F5)
Q16.R2 ↑	Strengthened: “record = BRST physical class” realized as state/observable duality on $A^0\{\text{BRST}, Z\}$ (Theorem Q16.F4).	DERIVED-CONDITIONAL (strengthened)

ID	Statement	Status
Q16.C2	Unchanged. NOT promoted by §20; remains conditional on $Z=\partial X$. The z^* -bridge (F5) reinforces but does not remove the postulate.	DERIVED-CONDITIONAL (on $Z=\partial X$)
E-Q16.2	Erratum to NC-Q16.10 (v2.2): its vector-space non-claim STANDS, but its clause “the OPENs are not closed” is updated — O-Q16.6’ and O-Q16.10 ARE closed in the correct (state-functor) category by §20. v2.2’s analysis was the correct diagnosis that located the category error.	ERRATUM (refinement).
NC-Q16.11	The closure of §20 is a state-functor equivalence, NOT a vector-space identity. The filter is NOT claimed isomorphic to $H^0(Q_BRST)$ as a vector space (that map fails, F-Q16.11). Theorem Q16.C2 is NOT promoted.	NON-CLAIM (active).
F-Q16.11	Falsification / wrong-category guard: the naive linear bijection $\Delta_1 \cong H^0$ (as a vector space) is exhibited as well-defined and structure-preserving, OR the state-functor map of F4 fails the commuting square.	PASSING: Δ_1 is not a vector space (a linear combination of states leaves the simplex); only the state/convex structure matches (§20 verification, Cat K).
O-Q16.11	Discharge the conditionality of Lemma Q16.F1 / Theorem Q16.F4 by deriving the identification of the BRST physical sector with the σ_z pointer basis from first principles — promoting F4 from DERIVED-CONDITIONAL to DERIVED.	OPEN.

§21. Operational Boundary Reconstruction and the Closure of O-Q16.11 [v2.4 increment]

This section, new in v2.4, closes the last record-layer OPEN, O-Q16.11 (discharge the σ_z -pointer / BRST-physical identification of Lemma Q16.F1). The strategy avoids the v2.3 residual head-on: it does NOT attempt to prove $Z = \partial X$ as an absolute topological boundary (ZS-Q12V §15 proves this cannot follow from mediation alone). Instead it reconstructs the **operational record boundary** $\partial_op X$ — the unique minimal-sufficient, holographically-saturated, BRST-physical Markov boundary of X — and identifies Z with it. A critical honest caveat is registered up front: this reconstruction closes the record/information layer, but it does NOT discharge the geometric conditionality of Theorem Q16.C2 (§21.7, NC-Q16.13).

§21.1 Lemma Q16.G1 — Dynamic Markov Separator

Lemma Q16.G1. Under $L_XY \equiv 0$ (ZS-F1, ZS-S1, PROVEN), X and Y are conditionally independent given Z : $X \perp Y \mid Z$. Equivalently, every $X \leftrightarrow Y$ transition factors through Z , with $\text{rank}(T_XY) = \text{rank}(V_ZY \vee V_XZ) \leq \dim Z = 2$ (ZS-Q7 Thm 2, DERIVED). By the structure theorem for quantum Markov chains (Hayden–Jozsa–Petz–Winter 2004), Z carries the full conditional-independence (recoverability) structure. Status: **DERIVED**. This is the dynamical-separation half identified in the v2.3 deep exploration, here stated as the operational Markov property.

§21.2 Lemma Q16.G2 — Minimality of the Z Separator

Lemma Q16.G2. Among separators carrying the binary pointer σ_z , Z is minimal: a boundary of $\dim < 2$ cannot carry a binary outcome ($\text{rank } 1 < 2$), and a boundary of $\dim > 2$ is not a minimal sufficient statistic for the binary X -record (it carries redundant degrees of freedom beyond the Fisher–Neyman minimal

sufficient statistic; Halmos–Savage 1949). Hence $\dim Z = 2$ is the unique minimal binary record boundary. Status: **DERIVED-CONDITIONAL** (given the rank-2 Kraus / binary-pointer structure of ZS-Q1 §3.3).

§21.3 Lemma Q16.G3 — Holographic Saturation

Lemma Q16.G3. The per-transit information ceiling is $I(N; J) \leq \ln(\dim Z) = \ln 2$, theorem-fixed (Holevo $\times$ ZS-Q7; §8 Cat C, locked to $\ln d$ across $d = 2 \dots 6$), and each Z -mediated transit deposits exactly $\Delta S = \ln 2$ (ZS-Q7 §6.3). Thus Z is not merely a separator but a **saturated record boundary**: one $\ln 2$ nat per boundary cell, matching the area law $S(\partial V) \leq |\partial V| \cdot \ln 2$ (ZS-Q6 Thm Q6.1). Status:

DERIVED-CONDITIONAL. Honest note: the area-law form of saturation imports the Regge boundary structure of ZS-Q6 (DERIVED-under-Regge), so this lemma is operational-plus-Regge, not purely information-theoretic.

§21.4 Lemma Q16.G4 — BRST Physical Boundary

Lemma Q16.G4. The physical record on Z is not the raw state but the BRST physical content: the J_Z -odd ghost / exact modes are quotiented out (Π_Z sandwich, ZS-M31.4; $\text{im } Q_B$ excluded), leaving the commutative pointer algebra $A^0_{\{BRST, Z\}} \cong \mathbb{C}^2$ of Lemma Q16.F1 (§20.1). Status:

DERIVED-CONDITIONAL (inherits the BRST-physical identification of v2.0/§20).

§21.5 Theorem Q16.G5 — Operational Boundary Reconstruction ($Z = \partial_{\text{op}} X$)

Theorem Q16.G5. Define the operational boundary $\partial_{\text{op}} X$ as the interface algebra satisfying (i) Markov separation, (ii) minimal sufficiency, (iii) holographic saturation, and (iv) BRST physicality. By Lemmas G1–G4, Z satisfies all four and is the unique such interface; therefore $Z = \partial_{\text{op}} X$. Status:

DERIVED-CONDITIONAL — more precisely DERIVED-under-operational-boundary-definition. This reconstructs the corpus’s record boundary from dynamics, information, and BRST structure, replacing the vague “geometric realization” of v2.3 with a definite operational object for the record layer. It is NOT a proof of an absolute topological boundary (NC-Q16.12).

§21.6 Theorem Q16.H1 — Seam Pointer = Center of the Boundary Algebra (closes O-Q16.11)

Theorem Q16.H1. Let $P_{\pm} = (1 \pm \sigma_Z)/2$. Then $P_+ P_- = 0$, $P_+ + P_- = 1$, each $P_{\pm}$ is BRST-closed and non-exact (the record outcomes are gauge-invariant and not ghost artifacts), and the J_Z -odd content vanishes under Π_Z (ZS-M31.4). Hence P_+ , P_- are the primitive idempotents of $A^0_{\{BRST, Z\}}$, and $\sigma_Z = P_+ - P_-$ is its CENTER generator. The seam pointer is therefore not an arbitrary choice but is forced as the center of the operational boundary algebra of Theorem Q16.G5. This DERIVES the σ_Z / BRST-physical identification that Lemma Q16.F1 had assumed, closing O-Q16.11. Status:

DERIVED-CONDITIONAL (DERIVED-under- ∂_{op} given G5).

§21.7 Critical Caveat, Status, Non-Claims, and Gate (v2.4)

Critical caveat (the integrity statement of v2.4). The operational reconstruction closes the record layer but does NOT promote Theorem Q16.C2. The 4π spin-closure that C2 ultimately rests on is derived (ZS-Q12V §14, Thm q12.bdy) from Z being the *topological* codimension-1 boundary of the 3D X -sector — which supplies the ambient $SO(3)$ frame and $\pi_1(SO(3)) = \mathbb{Z}/2$. The operational $\partial_{\text{op}} X$ is defined by information properties (Markov, minimal, saturating, BRST) and does NOT by itself supply that $SO(3)$

frame. Hence C2's residual is precisely the geometric embedding of $\partial_{\text{op}} X$ as the topological ∂X — the same residual identified in the v2.3 deep exploration — and it remains open.

Table 21.1. v2.4: O-Q16.11 closed by operational reconstruction; C2 NOT promoted (geometric residual explicit); two non-claims, one gate. The record/information layer is now fully closed; only the geometric embedding remains.

ID	Statement	Status
O-Q16.11 ↓	CLOSED via Theorems Q16.G5 & Q16.H1: σ_z is the center generator of the operational BRST boundary algebra $A^0_{\{\text{BRST}, Z\}} = \partial_{\text{op}} X$; the F1 identification is derived, not assumed.	OPEN → CLOSED (G5, H1)
$Z = \partial_{\text{op}} X$	The operational record boundary is reconstructed from Markov separation + minimal sufficiency + holographic saturation + BRST physicality (Theorem Q16.G5).	DERIVED-under-operational-boundary-definition
Q16.C2	NOT promoted. Remains conditional. Residual sharpened: the geometric embedding of $\partial_{\text{op}} X$ as the topological ∂X (the SO(3)-frame source of 4π), unchanged from v2.3.	DERIVED-CONDITIONAL (geometric residual)
NC-Q16.12	$Z = \partial_{\text{op}} X$ is an operational boundary reconstruction theorem, NOT an axiom-free proof of an absolute background topological boundary. The claim is that the record boundary is reconstructible from Markov separation, minimal sufficiency, holographic saturation, and BRST physicality.	NON-CLAIM (active).
NC-Q16.13	The operational reconstruction does NOT discharge the geometric 4π conditionality of Theorem Q16.C2. The information boundary $\partial_{\text{op}} X$ and the topological boundary ∂X coincide only under geometric realization (codim-1 embedding in 3D X with SO(3) frame), which remains the single residual postulate. C2 is NOT promoted.	NON-CLAIM (active).
F-Q16.12	Falsification: the σ_z eigenprojectors $P_{\pm}$ are shown to be BRST-exact (unphysical), OR a sufficient record boundary of $\dim \neq 2$ is exhibited, collapsing the minimal-binary reconstruction of Theorem Q16.G5.	PASSING: $P_{\pm}$ BRST-closed non-exact; $\dim Z = 2$ minimal (§21.2, §21.6; verification Cat L).

§22. Spin-Geometric Realization: The Geometric Residual, Decomposed [v2.5 increment]

This section, new in v2.5, examines a proposed Spin-Geometric Realization route to close the geometric residual of Theorem Q16.C2 — the embedding of the operational boundary $\partial_{\text{op}} X$ (§21) as a codimension-1 spin boundary of the 3D X -bulk, which is the SO(3)-frame source of the 4π closure (ZS-Q12V §14 Thm q12.bdy). The honest finding: the route **decomposes** the residual into three explicit, falsifiable sub-conditions, but it does NOT close it, because the three sub-conditions together are logically equivalent to the geometric embedding itself. Theorem Q16.C2 is therefore NOT promoted to an unconditional DERIVED; it remains **DERIVED-CONDITIONAL**, with its condition now decomposed rather than discharged.

§22.1 Theorem Q16.S0 — Operational-to-Regge Boundary Realization

Theorem Q16.S0. $Z = \partial_{\text{op}} X$ (Theorem Q16.G5) together with ZS-Q6 area-law saturation ($S(\partial V) \leq |\partial V| \cdot \ln 2$) and $\dim X = 3$ implies that the operational boundary admits a minimal 2D Regge-cell realization with area-cell entropy $\ln 2$. Status: **DERIVED-under-Regge**. Honest note: this theorem *inherits* the Regge boundary structure of ZS-Q6 (itself DERIVED-under-Regge); it does NOT derive that

structure. As such S0 is close to a restatement of ZS-Q6’s Regge boundary in operational language, not an independent advance toward the geometric embedding.

§22.2 Theorem Q16.S1 — Regge Boundary Spin-Lift

Theorem Q16.S1. A 2D oriented Regge boundary OF a 3D X-bulk carries the ambient $SO(3)$ frame (two tangent + one normal direction); its spin lift is $Spin(3) = SU(2)$, and via $\pi_1(SO(3)) = \mathbb{Z}/2$ the boundary spinor returns only after 4π (2π gives the sign flip). Status: **IMPORTED** ($Spin(3) = SU(2)$ and the $SU(2) \rightarrow SO(3)$ double cover are standard; Milnor–Stasheff 1974; already verified in zs_q16_verify Cat H). Critical caveat: S1 ASSUMES the codim-1 embedding — the phrase “a 2D boundary OF a 3D bulk” is precisely the embedding in question. Moreover, on a bare oriented 2-surface the spin structure is not forced but is a CHOICE (an $H^1(\Sigma; \mathbb{Z}/2)$ -torsor; a 2π non-spinor structure is equally consistent), and it is forced to the spinor (4π) lift only by the ambient $SO(3)$ frame of the embedding (ZS-Q12V §14 Thm q12.dim2, PROVEN). Thus S1 reproduces Thm q12.bdy; it does not derive the embedding the 4π closure needs.

§22.3 The Decomposition Is Not a Reduction

Combining S0 and S1, the condition “ReggeSpinBoundary” names three sub-conditions: (i) a 2D Regge realization of the boundary (from ZS-Q6, conditional on Regge), (ii) a normal-bundle reconstruction (a distinguished outward normal), and (iii) oriented-frame closure ($T(\partial_{\text{op}} X) \oplus N$ reconstructing the rank-3 $SO(3)$ frame, with a spin structure). But (i)+(ii)+(iii) jointly state exactly that Z is a codimension-1 spin submanifold of the 3D X-bulk — which is **logically equivalent** to the geometric postulate $Z = \partial X$. Hence ReggeSpinBoundary is a **clarifying decomposition** of C2’s residual into named, individually-falsifiable pieces, NOT a reduction to anything more primitive. The residual is sharper and more analyzable, but it is the same residual.

§22.4 The One Genuine Lead — Normal Bundle from Record Flow (O-Q16.12)

Of the three sub-conditions, only (ii) the normal-bundle reconstruction is a candidate for genuine derivation: the X-side record flow — the directed information transfer $X \rightarrow Z \rightarrow Y$ — might supply a distinguished outward normal, so that $T(\partial_{\text{op}} X) \oplus N \sqsubset$ reconstructs the $SO(3)$ frame from information data (an information $\rightarrow$ geometry bridge). The obstruction: for the record flow to be a *geometric* normal (a spatial direction in 3D X) rather than an abstract information arrow, the X-sector must already be a 3D spatial manifold and the flow spatially realized — which presupposes the very geometric structure at issue. This connects the residual to the external emergent-spacetime program (deriving spatial geometry from entanglement/information), and is registered as O-Q16.12. It is the genuine, and external, residual.

§22.5 Status, Corollary, Non-Claim, and Gate (v2.5)

Table 22.1. v2.5: C2’s geometric residual decomposed (S0, S1, S2) but NOT closed; C2 stays DERIVED-CONDITIONAL; one non-claim, one external OPEN, one gate. The genuine residual is the information $\rightarrow$ geometry (emergent-spacetime) bridge.

ID	Statement	Status
Q16.S2 (Cor.)	Given S0 and S1, Theorem Q16.C2’s conditionality is re-expressed: from the bare postulate $Z=\partial X$ to the decomposed ReggeSpinBoundary (Regge realization + normal-bundle + oriented-frame/spin). This is a re-expression of the condition, not a discharge of it.	C2 condition decomposed (NOT promoted)

ID	Statement	Status
Q16.C2	NOT promoted. Remains conditional. Following Criterion 2 (whole-corpus foundations): C2 stays DERIVED-CONDITIONAL, its condition reduced/decomposed to ReggeSpinBoundary rather than raw $Z=\partial X$. It is NOT an unconditional / axiom-free DERIVED.	DERIVED-CONDITIONAL (condition decomposed)
NC-Q16.14	The Spin-Geometric Realization DECOMPOSES C2's geometric residual into named sub-conditions; it does NOT close it. "ReggeSpinBoundary" is logically equivalent to the geometric embedding $Z=\partial X$, so C2 is NOT promoted to an unconditional DERIVED. S0 inherits (does not derive) the ZS-Q6 Regge structure; S1 assumes the codim-1 embedding and the spin structure is a choice forced only by that embedding.	NON-CLAIM (active).
O-Q16.12	Derive the ReggeSpinBoundary — specifically the normal-bundle (and hence the codim-1 embedding) — from the action / record-flow information structure alone, promoting C2's condition from assumed to derived. Connects to the external emergent-spacetime program.	OPEN (external).
F-Q16.13	Falsification: the X-side record flow is shown NOT to define a consistent outward normal (no information→geometry bridge), OR a 2π (non-spinor) structure on the physical Z-boundary is exhibited, collapsing the 4π closure (cf. F-q12.5).	Open test; tied to O-Q16.12.

Acknowledgements and Code Availability

This work was developed under Kenny Kang's editorial direction with AI assistance (Anthropic Claude) for derivation cross-checking, simulation, and drafting; the author assumes full responsibility for all scientific content. The verification script `zs_q16_outcome_mc_v1_0.py` (numpy only, canonical seed 20260604) reproduces every numerical result of §8: execution "python3 zs_q16_outcome_mc_v1_0.py", expected 27/27 PASS.

Appendix A. The Repeated-Indirect-QND Filter Model

A system with N pointer eigenstates is probed by a dim- d probe; $p(j | n)$ is the probability of probe outcome j given eigenstate n . The conditional state is summarized by the posterior q over $\{n\}$, updated by Bayes' rule $q_n \leftarrow q_n p(j | n) / \sum_m q_m p(j | m)$ on the realized outcome j , which is sampled from the marginal $\sum_n q_n p(j | n)$. By the equivalence of the predictive and mixture laws, generating the record from the running marginal is statistically identical to fixing a "true" eigenstate $n^* \sim q^{(0)}$ (Born) and drawing the record i.i.d. from $p(\cdot | n^*)$; the latter is used for speed. The pointer probabilities form a martingale; under non-demolition and irreducibility (no dark subspace), $q \rightarrow \delta_{\{n^*\}}$ almost surely with n^* Born-distributed (T1–T4). The symmetric probe used in §8 sets $p(\cdot | n)$ to tilt outcome $(n \bmod d)$ by a strength $\varepsilon \in [0, 1]$; $\varepsilon = 0$ is uninformative, $\varepsilon \rightarrow 1$ is projective and saturates (2).

Appendix B. Estimation Stability — Formal Statement

Let $(\Omega, \mathcal{F}, P)$ carry the measurement record. For the QND stochastic master equation with generator fixed by the probe channel, let $\rho^\wedge(t)(q)$ denote the conditional state at time t started from prior q and driven by the realized record ω . Theorem (Benoist–Pellegrini [4]; discrete: Bauer–Benoist–Bernard [3]): if the measurement is non-demolition and the associated quantum channel is irreducible (no dark subspace), then for any two full-support priors q, q' , $\|\rho^\wedge(t)(q) - \rho^\wedge(t)(q')\| \rightarrow 0$ P -almost surely, exponentially fast, and the common limit is the random pure projector selected by ω , with selection probability given by the

Born weights of q for the true law. The §8 Block 3b simulation is a finite-time numerical instance with three priors, including an adversarial one.

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Version History

v2.5 (June 2026): Strictly additive over v1.0–v2.4 (no deletion). Adds §22, examining a Spin-Geometric Realization route to close Theorem Q16.C2’s geometric residual. Honest outcome: the route DECOMPOSES the residual but does NOT close it. Theorem Q16.S0 (Operational-to-Regge, **DERIVED-under-Regge**) inherits — does not derive — the ZS-Q6 Regge boundary structure. Theorem Q16.S1 (Regge Boundary Spin-Lift, **IMPORTED**, Milnor–Stasheff 1974) supplies $\text{Spin}(3) = \text{SU}(2) \Rightarrow 4\pi$ but only after assuming the codim-1 embedding, and on a bare surface the spin structure is a CHOICE forced only by that embedding (ZS-Q12V Thm q12.dim2). Corollary Q16.S2 re-expresses C2’s condition as the decomposed ReggeSpinBoundary (Regge realization + normal-bundle + oriented-frame/spin), which is logically equivalent to the geometric embedding $Z=\partial X$ — a clarifying decomposition, NOT a reduction. Following Criterion 2 (whole-corpus foundations), Theorem Q16.C2 is NOT promoted: it remains **DERIVED-CONDITIONAL**, its condition decomposed rather than discharged. The one genuine lead — reconstructing the normal bundle from the X-side record flow (information→geometry) — is registered as O-Q16.12 and connects to the external emergent-spacetime program. Adds Theorems Q16.S0–S1, Corollary Q16.S2, NC-Q16.14, O-Q16.12, F-Q16.13, and two references (Regge 1961; Milnor–Stasheff 1974). The §22 spin-lift facts are already verified in zs_q16_verify Cat H. Zero new free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

v2.4 (June 2026): Strictly additive over v1.0–v2.3 (no deletion). Adds §21, closing the last record-layer OPEN, O-Q16.11, by an Operational Boundary Reconstruction. Lemmas Q16.G1 (Markov separation $X \perp Y | Z$, **DERIVED**, ZS-Q7 Thm 2 + Hayden–Jozsa–Petz–Winter 2004), G2 (minimal binary boundary, **DERIVED-CONDITIONAL**, Halmos–Savage 1949), G3 (holographic saturation $I \leq \ln 2$ per cell, **DERIVED-CONDITIONAL**, operational-plus-Regge), G4 (BRST physical boundary $A^0_{\{\text{BRST}, Z\}}$, **DERIVED-CONDITIONAL**); Theorem Q16.G5 ($Z = \partial_{\text{op}} X$, **DERIVED-CONDITIONAL** / **DERIVED-under-operational-boundary-definition**); Theorem Q16.H1 (σ_Z is the CENTER generator of $A^0_{\{\text{BRST}, Z\}}$, **DERIVED-CONDITIONAL**, closing O-Q16.11). CRITICAL: Theorem Q16.C2 is NOT promoted — the operational $\partial_{\text{op}} X$ does not supply the geometric $\text{SO}(3)$ -frame of the 4π closure (ZS-Q12V §14), so C2 remains **DERIVED-CONDITIONAL** with its residual the geometric embedding of $\partial_{\text{op}} X$ as the topological ∂X (NC-Q16.13). Adds NC-Q16.12 (operational reconstruction, not absolute topological proof), NC-Q16.13 (C2 geometric residual undischarged), F-Q16.12, and two references

(Hayden–Jozsa–Petz–Winter 2004; Halmos–Savage 1949). The record/information layer (O-Q16.6', O-Q16.10, O-Q16.11) is now fully closed; only the geometric embedding remains. Zero new free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

v2.3 (June 2026): Strictly additive over v1.0–v2.2 (no deletion). Adds §20, closing the two problems v2.2 left CONFIRMED-OPEN by correcting the category. The v2.2 obstruction (filter continuum vs cohomology vector space) was a category error; the correct object is the state functor on the BRST physical pointer algebra $A^0_{\{\text{BRST}, Z\}} \cong \mathbb{C}^2$, whose state space $\text{State}(\mathbb{C}^2) = \Delta_1$ is itself a continuum. Lemma Q16.F1 (BRST pointer algebra $\cong \mathbb{C}^2$, **DERIVED-CONDITIONAL**), Lemma Q16.F2 (filter-as-state, **PROVEN**, Gelfand–Naimark 1943), Lemma Q16.F3 (Bayes = BRST conditioning, **PROVEN** / DERIVED-CONDITIONAL, Ozawa 1985), Theorem Q16.F4 (full functorial bijection $\text{Filt}_Z \cong \text{State}(A^0_{\{\text{BRST}, Z\}})$, **DERIVED-CONDITIONAL**, closing O-Q16.6'), Theorem Q16.F5 (z^* -collapse fixed points $\leftrightarrow$ primitive characters / Z-seam cohomology classes, **DERIVED-CONDITIONAL** on $Z=\partial X$, closing O-Q16.10). Proposition Q16.R2 strengthened to DERIVED-CONDITIONAL (state/observable duality). Theorem Q16.C2 NOT promoted — remains DERIVED-CONDITIONAL on $Z=\partial X$. Adds Erratum E-Q16.2 (refines NC-Q16.10), NC-Q16.11 (state-functor equivalence, not vector-space identity), F-Q16.11 (wrong-category guard), O-Q16.11 (discharge F1 condition), and two references (Gelfand–Naimark 1943; Ozawa 1985). Verified by `zs_q16_verify_v2_3.py` Category K. Zero new free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

v2.2 (June 2026): Strictly additive over v1.0–v2.1 (no deletion). Adds §19, a five-step parallel deep exploration attempting to close O-Q16.9 (collapse = z^* -convergence without the 4π axiom $\rightarrow$ C2 DERIVED) and O-Q16.6' (filter = BRST cohomology bijection $\rightarrow$ R2 DERIVED). Honest outcome: **CONFIRMED-OPEN** for both — neither closes with current corpus tools, and no promotion of C2 or R2 is made (NC-Q16.10). Two clarifications: (i) the 4π -closure is DERIVED from $Z=\partial X + (Z, X)=(2,3)$ (ZS-Q12V §14 Thm q12.bdy), so Theorem Q16.C2's conditionality relocates to the single corpus postulate $Z=\partial X$ (not a separate 4π axiom); (ii) O-Q16.6''s obstruction sharpens to a two-cohomology mismatch — the filter carries an information-cohomology structure (Baudot–Bennequin 2015; Vigneaux 2020) distinct from BRST cohomology, with a continuum-vs-finite-dimensional dimension gap (ZS-M27 §7). New structural observation: both OPENS meet at z^* (the filter's collapse fixed points $\leftrightarrow$ the BRST classes), registered as O-Q16.10. Adds NC-Q16.10, O-Q16.10, and two references (Baudot–Bennequin 2015; Vigneaux 2020). Verification 27/27 PASS unchanged. Zero new free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

v2.1 (June 2026): Strictly additive over v1.0–v2.0 (no deletion). Adds §18, consolidating a five-step deep exploration. Registers Theorem Q16.C2 (**DERIVED-CONDITIONAL**): the per-cycle J_Z -odd leak $1 - |\lambda|^2 = 0.2052$ is the damping that makes z^* an attracting fixed point (ZS-M1 Remark 1.2; Koenigs 1884: attracting requires $0 < |f'| < 1$) and is therefore a necessary condition for wave-function collapse (= convergence to z^* , ZS-Q12.A) and the microscopic face of the arrow of time (ZS-Q7); gauge-redundant (BRST-exact, no observable information) yet functionally essential — not waste. Adds NC-Q16.9 (no observable information $\neq$ no function; not waste / not an informational reference / not an archive), F-Q16.10 (falsification: collapse with $|\lambda| = 1$, or the leak unnecessary for convergence), O-Q16.9 (remove the 4π -closure conditionality to promote C2 to DERIVED), and one reference (Koenigs 1884). The §18.4 80/20 reading: the definite 79.5% output cannot exist without the 20.5% damping engine. Verification 27/27 PASS unchanged. Zero new free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

v2.0 (June 2026): Strictly additive over v1.0–v1.2 (no deletion). Adds §17, consolidating a five-step deep exploration of the J_Z-odd register slot (slot 1). Two status upgrades: Theorem Q16.C1 (fine/coarse consistency) **DERIVED-CONDITIONAL** → **DERIVED**, its environment-identification condition discharged because slot 1 is BRST-exact (ZS-M22 §6.6.4), decouples from X/Y (ZS-F0 §8.8), and is observable-invisible (ZS-M31.4), with BRST-exact $\Rightarrow$ null/unphysical (Kugo–Ojima 1979; Henneaux–Teitelboim 1988/1992); Proposition Q16.R2 (Cobordism Ledger) **HYPOTHESIS-strong** → **DERIVED-CONDITIONAL**, grounding “record = BRST cohomology class $H^0(Q_BRST)$ ” with R1/R2 agreeing on the ghost exclusion (residual bijection O-Q16.6'). Registers the “Backbone” reading: observational invisibility and ontological preservation **DERIVED**, but “accumulating archive” is **NON-CLAIM** (NC-Q16.8, conflicts with R4). Adds O-Q16.6', O-Q16.8, and two references (Kugo–Ojima 1979; Henneaux–Teitelboim 1988/1992). Verification 27/27 PASS unchanged. Zero new free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

v1.2 (June 2026): Strictly additive over v1.0 and v1.1 (no deletion). Documents Erratum E-Q16.1: the §15.4 wording reading the per-cycle Wilson contraction $|\lambda|^2 = 0.7948$ as ~20% information “forgotten” is corrected — by ZS-F0 Thm 12.1–12.3 it is a partition-function survival probability whose complement 0.2052 is transferred to the J_Z-odd register mode (sum rule ≈ 1), with the time evolution $U(t) = e^{\{-iHt\}}$ a separate unitary object (Thm 12.2, 12.4), so global information is conserved. Adds §16 (Theorem Q16.C1, **DERIVED-CONDITIONAL**): the fine-grained transfer and the coarse-grained record forgetting (ZS-Q7 $\Delta S = \ln 2$) are one process at two resolutions, related by the Stinespring (1955) / GKSL (Lindblad 1976) reduction with the J_Z-odd ghost mode as the traced-out environment; §16.1 corrects the “20.5% → Y-sector” reading (the transfer is to the J_Z-odd mode, only the 0.0001 X–Y residual touches Y). Adds NC-Q16.7, F-Q16.9, and three references (Lindblad 1976; Gorini–Kossakowski–Sudarshan 1976; Stinespring 1955). Verification ledger unchanged (27/27 PASS). Zero new free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

v1.1 (June 2026): Strictly additive increment over v1.0 (no deletion). Adds §15 (“How the Universe Records Reality: Four Principles”) consolidating a five-step deep exploration of the corpus recording mechanism: Theorem Q16.R1 (Record Compression, **DERIVED-CONDITIONAL** via Zakai / Kushner–Stratonovich filtering — the present conditional Z-record state is a sufficient statistic); Proposition Q16.R2 (Cobordism Ledger, **HYPOTHESIS-strong** — the record is the gauge-invariant cohomology class of the boundary state under the BV-BFV mQME / $\mathcal{H}_{cob}$; the “boundary-state-vs-cohomology” dichotomy is dissolved); Principle Q16.R3 (X-Single / Y-Many Separation, operational **DERIVED-CONDITIONAL** / ontological many-worlds **NON-CLAIM**); Principle Q16.R4 (No-Archive Reality, **DERIVED-CONDITIONAL**). Adds F-Q16.7–8, NC-Q16.6, O-Q16.6–7 and four references (Zakai 1969; Stratonovich/Kushner; Bain–Crisan; Cattaneo–Mnev–Reshetikhin / Cattaneo–Moshayedi–Wernli). Verification ledger unchanged (27/27 PASS; additions are theorem-grounded / IMPORTED). Zero new free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; consciousness firewall intact.

v1.0 (June 2026): Initial public release. Resolves the status of NC-F19.7 by the three-level decomposition (weak / strong / absolute). Weak sufficiency **DERIVED-CONDITIONAL** via imported QND convergence theorems [2–4] reinterpreted with z^* and the $L_{XY} \equiv 0$ purification condition; rate ceiling $\ln(\dim Z)$ **DERIVED-CONDITIONAL** via Holevo $\times$ ZS-Q7. Strong sufficiency **OPEN**, sharpened to non-epistemic via estimation stability [3,4] (§7). Absolute-strongest form **NON-CLAIM**. Six falsification gates (F-Q16.1–6), five non-claims (NC-Q16.1–5), five OPEN gates (O-Q16.1–5). Verification 27/27

PASS (zs_q16_outcome_mc_v1_0.py, seed 20260604); cross-paper consistency 9/9. (A, Q, dim Z) = (35/437, 11, 2) LOCKED; zero new free parameters. Consolidated from internal Z-Spin Collaboration research notes following ZS-Q15 v1.2.